When Are Robust-RAG Results Comparable? A Coordinate Audit of Published Comparisons

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Abstract

Robust retrieval-augmented generation (RAG) papers share labels such as noise robustness, faithfulness, security, and calibration while answering different decision questions. We audit twelve cross-paper comparisons that published robust-RAG papers either make directly or invite by grouping, using five coordinates: evidence state, origin/capability, intervention, outcome, and policy/utility. Ten do not survive as stated. In the sharpest case, a quoted 3.7-point attack-success lead reverses to -45.3 points once the two sides are read at a matched insertion budget, using an ablation reported in the attacking paper itself. Two of three within-harness contrasts pass unchanged, so the audit is not vacuous. Checking such claims is itself a measurement problem. Of 39 venue-linked papers, 9 collect original human judgments; 2 report how many annotators, and none reports the item–annotator assignment or releases pre-aggregation votes. That gap has a cost we can price: reviewer heterogeneity is estimable only from a crossed design, and at the value we estimate ($\hat{\sigma}_r = 0.27$) a source-only test of panel-majority correctness rejects a true null 24.2% of the time at 80 sources, rising with more sources, whereas two analyses computed at the reviewer level hold 5% and retain power. Both are executable retrospectively from votes no audited paper releases. We release source-linked cases, a verifier, model outputs, and all simulations.

1 Introduction

Retrieval-augmented generation conditions a generator on retrieved evidence (Lewis et al., 2020; Karpukhin et al., 2020; Guu et al., 2020), and robust-RAG studies ask what happens when that evidence is irrelevant, insufficient, conflicting, stale, weakly sourced, or malicious. Two reported "robustness" gains may nevertheless concern different retrieval snapshots, attacker capabilities, interventions, outcome units, or deployment actions. Source verification is therefore necessary but insufficient: a cross-paper claim must also establish that both results answer a compatible decision question.

An answer, in one case. PoisonedRAG reports a 97% black-box attack success rate on Natural Questions by injecting five malicious texts per target question (Zou et al., 2025). Phantom reports 93.3% success for its refusal objective using a single poisoned document (Chaudhari et al., 2024). Placed side by side, PoisonedRAG leads by 3.7 points. The two numbers are measured at different attacker budgets, and PoisonedRAG's own ablation gives its black-box rate when one of the five retrieved texts is malicious: 0.48. At a matched budget the 3.7-point lead becomes a 45.3-point deficit. Nothing was reproduced and nothing was mis-cited; a single unstated coordinate reversed the ordering. This is the kind of failure the paper detects, quantifies, and—where the sources permit—repairs.

What the audit answers. Two robust-RAG results are comparable when they agree on five coordinates: the evidence state (C1), the origin and capability envelope that produced it (C2), the intervention (C3), the measured outcome and its unit (C4), and the action set and loss (C5). We apply this to twelve comparisons drawn from published robust-RAG papers and surveys and find that ten do not survive as stated: six have no common scope at all and four survive only on a strict subset. Of three within-harness contrasts, in which one paper compares systems under its own corpus, retriever, generator and metric, two pass unchanged, so the audit distinguishes rather than condemns. Every reported value is copied from a locally extracted full text and re-verified at its source line by a released checker.

Checking claims is itself a measurement. If a claim-review prompt is assigned to six people but hundreds of their labels are analyzed only

ID	Compared results	Quoted gap	Mismatch	Verdict	Consequence of matching the coordinate
SEC-01	PoisonedRAG vs. CorpusPoisoning	+0.96 ASR	C1, C2, C4	undefined	gap 0.96→0.47 at matched budget; but C4 mismatch (retrieval hit rate vs answer ASR) remains; comparison still undefined
SEC-02	PoisonedRAG vs. Phantom	+3.7% ASR in favor of PoisonedRAG	C1, C2, C4	undefined	gap +3.7% → -45.3%; ranking reverses at matched budget, but C4 (different objectives) makes the comparison undefined
SEC-03	RobustRAG vs. PoisonedRAG	Defense reduces ASR from 0.97 to <=0.15	C1, C2, C4	narrows	defense certified at k'=1 where attack ASR=0.48, not 0.97…
SEC-04	BadRAG vs. PoisonedRAG	+1.2% in favor of BadRAG (98.2% vs 97%)	C2, C4, C5	undefined	gap +1.2% is uninterpretable; different objectives (DoS vs targeted answer) preclude ranking
SEC-05	AIP vs. PoisonedRAG	+66.66% ASR in favor of AIP	C1, C2	narrows	gap +66.66% survives on MedSquad at matched budget; but domain transfer (NQ vs MedSquad) limits generalizability
NOISE-01	CRAG (Self-CRAG) vs. Self-RAG 7B	+6.9 accuracy in favor of Self-CRAG	C3	survives	gap +6.9 points → unchanged; coordinates match
NOISE-02	CtrlA 7B vs. Self-RAG 7B	+6.9 accuracy in favor of CtrlA	C1, C3	narrows	gap +6.9 points has unknown sign after controlling for backbone/retriever difference
NOISE-03	RGB (ChatGPT) vs. RAAT (ChatGPT3.5)	+4.53 accuracy/EM in favor of RGB's ChatGPT	C1, C4	undefined	gap +4.53 has unknown sign: substring accuracy >= strict EM, so RGB's advantage partly or wholly reflects metric…
NOISE-04	Chain-of-Note vs. RAAT (ChatGPT3.5)	+7.55 EM (ChainOfNote) vs +1.68 EM (RAAT)…	C1, C3, C4	undefined	improvements +7.55 vs +1.68 are incomparable: ChainOfNote's baseline (34.28) operates without gold context vs RAAT's…
NOISE-05	AstuteRAG vs. RobustRAG	+4.4 accuracy in favor of ASTUTE RAG	C1, C5	survives	gap +4.4 points → unchanged; coordinates match for accuracy comparison
FAITH-02	FaithfulRAG vs. FaithEval (GPT-4o CF)	FaithfulRAG 81.7% vs GPT-4o 47.5% (+34.2% on…	C1, C4	undefined	gap 81.7% vs 47.5% conflates aggregate vs single-task, different conflict types, and different models…
CALIB-03	ADAPT-LLM vs. Self-RAG 7B	+16.75 accuracy in favor of Self-RAG	C1, C4	narrows	gap +16.75 confounds harder test subset + stricter metric for ADAPT-LLM; true method difference is unknown
Within-harness controls (same paper, same retriever, generator, corpus and metric)					
FAITH-01	RAGTruth (resp-level) vs. RAGTruth (span-level)	Response-level F1 78.7% vs span-level (not…	C4	narrows	response-level F1 78.7% cannot be compared to span-level metrics; C4 granularity mismatch
CALIB-01	Self-RAG 7B vs. RAG-baseline	+23.9 accuracy in favor of Self-RAG	C3	survives	gap +23.9 points → unchanged; within-paper controlled experiment
CALIB-02	CtrlA 7B vs. FLARE (reimpl.)	+4.0 accuracy in favor of CtrlA	C3	survives	gap +4.0 points → unchanged; properly controlled comparison

Table 1: Worked cross-paper comparisons from the robust-RAG literature. Of 12 comparisons that published papers either make directly or invite by grouping, 6 have no common scope, 4 survive only on a strict subset, and 2 survive as stated; 3 within-harness contrasts act as negative controls. Every number is copied verbatim from a locally extracted full text and re-checked at the cited line range by `verify_cross_paper_cases.py`. Coordinates are C1 evidence state, C2 origin/capability, C3 intervention, C4 outcome, C5 policy/utility. Appendix Table 11 gives the source lines and the coordinate-matched values.

by item or source, the apparent sample size concerns repeated judgments, not treatment assignments. These consequences follow from crossed participant–item designs, clustering, and reference construction (Baayen et al., 2008; Barr et al., 2013; Abadie et al., 2023; Card et al., 2020). We make them operational for robust-RAG claim review, and we show that the reporting practice needed to act on them is absent. Of 39 venue-linked papers with local full text, 9 collect original human judgments; 2 report the number of annotators, none reports the item–annotator assignment, and none releases pre-aggregation votes.

That gap has a measurable price. Between-reviewer heterogeneity is identified only by a crossed reviewer-by-item design; from the one such design available to us we estimate $\hat{\sigma}_r = 0.27$ on the logit scale, and we show that the agreement statistics published papers do report place no usable restriction on it. At that value, the analysis these papers make possible—a source-level test of panel-majority correctness—rejects a true null 24.2% of the time at 80 sources and 36.7% at 160, while two reviewer-level analyses hold the nominal 5% and still detect a real prompt effect. Both remedies are executable after data collection, but only from votes that no audited paper releases. The reporting gap and the design failure are therefore the same finding seen twice.

Contributions. (i) A five-coordinate comparability audit applied to twelve real published comparisons, with verbatim numbers, coordinate-matched re-readings, and a verifier (§4). (ii) A report-

ing audit of the nine robust-RAG papers that collect human judgments, positioned against the NLG human-evaluation literature (§5). (iii) A consequence-defined benchmark with a length-matched placebo arm and a pre-declared validity gate (§6). (iv) An empirically anchored simulation identifying two retrospectively executable analyses that restore calibration without losing power (§7). We also report one negative result: used as a four-way classifier on an 80-claim development sample, no variant of the schema separates from predicting the modal verdict (§5.1). The coordinates are disclosure fields validated on published comparisons, not a human coding instrument.

2 Positioning

RAG surveys and shared evaluation. Architectures differ in where retrieval enters and how evidence is fused (Chen et al., 2017; Khandelwal et al., 2020; Izacard and Grave, 2021; Borgeaud et al., 2022; Izacard et al., 2023; Shi et al., 2024). General surveys organize these and their applications, while evaluation, trustworthiness, security, conflict, and abstention surveys expose different failure surfaces (Gao et al., 2024; Yu et al., 2024; Zhou et al., 2024; Xu et al., 2026, 2024; Wen et al., 2025). Common-harness comparisons improve empirical control (Wang et al., 2024; Alushi et al., 2026), but they do not resolve whether a quoted cross-paper result concerns the same evidence state, capability envelope, outcome, or action. Our unit of analysis is not a paper or a benchmark but an individual cross-paper juxtaposition, and our output is a per-comparison verdict with a coordinate-matched number where one is recoverable. Appendix Table 19 positions this lens.

Human-evaluation reporting. Auditing how human evaluations are reported has an established literature. Howcroft et al. (2020) examine 165 NLG papers and document exactly this class of under-reporting; van der Lee et al. (2021) survey practice and issue best-practice guidelines; Belz et al. (2021) and Belz et al. (2023) show that previous human evaluations frequently cannot be reproduced or even assessed. Clark et al. (2021) show that untrained evaluators fail at the judgments they are asked to make, and Bender and Friedman (2018) argue for structured dataset disclosure. Our reporting audit is smaller and single-coded, and it makes no prevalence claim that these studies do not already establish more strongly. Its increment is narrower and specific: it asks, for robust-RAG papers only, whether the fields required to *reanalyze* a reported human evaluation at the assignment unit are present, and it prices the consequence of their absence in §7. Card et al. (2020) make the canonical NLP argument for powering the randomized unit, and van Miltenburg et al. (2021) the case for preregistration; our contribution there is not the principle but the executable demonstration of which retrospective analyses recover it.

Human judgments as measurements. Annotation research distinguishes reproducibility from truth (Artstein and Poesio, 2008), and observer-error models predate its use in NLP (Dawid and Skene, 1979). Disagreement is often signal rather than noise (Uma et al., 2021). Majority voting can erase structured variation (Plank, 2022; Davani et al., 2022), while annotator models retain rater and item effects (Passonneau and Carpenter, 2014; Paun et al., 2018). Crossed participant–item designs require both sources of variation (Baayen et al., 2008), and clustering follows treatment assignment rather than row count (Abadie et al., 2023).

3 The Two-Layer Audit

Four recurring boundaries in the robust-RAG literature motivate the coordinates. Evidence availability is not evidence use, so a context filter, a noise-trained reader, and a diagnostic benchmark do not rank interchangeably (Yoran et al., 2024; Fang et al., 2024; Chen et al., 2024a; Yu et al., 2023). Support, authority, and conflict are distinct, so a response can be correct but unsupported or faithful to a superseded source (Niu et al., 2024; Wang et al., 2025a; Vu et al., 2024). Natural retrieval failure and adaptive attack are different games played under different access, budget, and persistence assumptions (Zou et al., 2025; Chaudhari et al., 2024; Xiang et al., 2024). And a confidence signal is not an action policy: comparing abstention or adaptive-retrieval methods needs a common action set, budget, and loss (Guo et al., 2017; Geifman and El-Yaniv, 2017; Asai et al., 2024; Yan et al., 2024; Joren et al., 2025). Appendix A develops these with paper-level contrasts.

The substantive layer accordingly records five coordinates for each side of a comparison: evidence state, origin/capability, intervention, outcome, and policy/utility, followed by claim-specific controls (Appendix Table 20). These coordinates

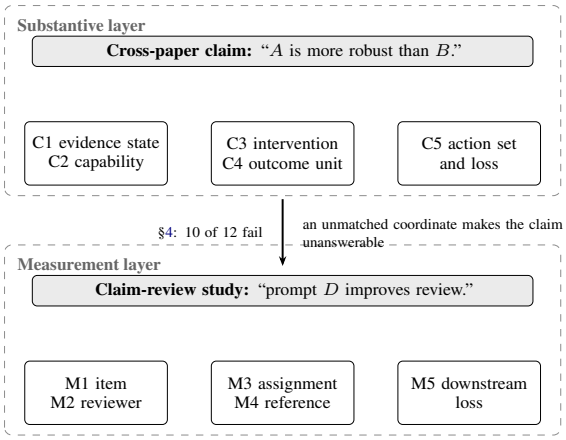


Figure 1: The two layers answer different questions about the same sentence. The substantive layer asks whether two published results concern one decision problem; Table 1 applies it to real comparisons. The measurement layer asks what a study of claim review estimates; Table 3 shows which analyses of such a study are valid. Passing one layer cannot repair failure in the other.

specialize existing distinctions from evidence synthesis, selective prediction, and security evaluation (Richardson et al., 1995; Chow, 1970; Papernot et al., 2018). The measurement layer then asks how judgments about those coordinates were produced: which people, items, assignments, references, and downstream losses define the target. Passing one layer cannot repair failure in the other (Figure 1).

Decision rule. For a proposed comparison, record each coordinate on both sides from the sources alone. If every coordinate matches, the quoted gap stands. If a coordinate differs but one side reports an operating point at the other side's setting, re-read the comparison at that matched point and report the matched number. If a coordinate differs and no matched operating point exists, the gap is undefined and the comparison must be replaced by the common scope both studies support, or dropped. §4 applies exactly this rule.

Three targets. A majority-label difference for two realized reviewer panels is a *panel contrast*. A prompt effect over reviewers instead averages individual potential judgments over a defined participant and item population. A claim of better decisions requires a third target: expected downstream loss for accepting, narrowing, rejecting, or deferring a comparison. These targets coincide only under assumptions that should be stated rather than inherited from the aggregation rule. Appendix Section F gives the notation.

Assignment and reference. Source resampling with reviewers fixed measures item variation conditional on the realized panel; if prompts are assigned to reviewers, reviewer-composition variation remains as claims accumulate, so more labels per person improve conditional precision without creating new prompt assignments. Separately, blinding reference writers to arm responses prevents copying, but a rubric that paraphrases the tested prompt still embeds the intervention. We therefore say "reference-relative agreement," reserving "accuracy" for an intervention-independent criterion.

4 Twelve Published Comparisons

We assembled comparisons of two kinds from 259 locally extracted full texts: *direct* (eight cases), where one paper reports its own number beside a number attributed to another paper, and *survey-invited* (four), where a survey or benchmark groups two systems under one label so that a reader would rank their headline numbers. Every value is copied verbatim with a file and line range, and `verify_cross_paper_cases.py` reopens each range and fails on any mismatch; 70 checks pass. Ten of twelve do not survive as stated: six have no common scope and four hold only on a strict subset. The most frequently violated coordinate is the evidence state (10 of 12), followed by the outcome unit (8) and origin/capability (5). We also audited three within-harness contrasts, in which one paper compares systems under its own corpus, retriever, generator and metric; two pass unchanged and one narrows (Appendix Table 11).

Capability envelopes decide security comparisons. All five security cases involve an unmatched attacker capability or objective. Beyond the budget reversal in §1, PoisonedRAG's 97% is also quoted against a corpus-poisoning attack whose reported quantity is a retrieval hit rate rather than an answer-level success rate, so matching the budget alone moves the gap from 0.96 to 0.47 without making the numbers commensurable. In the defense direction, RobustRAG's certificate is stated at a corruption budget of one passage, where the attack it is quoted as neutralizing succeeds 48% rather than 97%: "reduces attack success from 97%" overstates the threat the certificate covers. AIP's advantage over PoisonedRAG narrows rather than dissolves, surviving at a matched budget on

the shared domain only.

Outcome units decide noise comparisons. "Noise robustness" names at least two measurements. RGB scores ChatGPT at 76.00% under a substring-match accuracy at a stated noise ratio; RAAT scores ChatGPT-3.5 at 71.47 exact match under a mixture of three constructed noise types with the gold passage always present. The apparent 4.53-point gap combines a difference in metric strictness with a difference in what counts as noise, and neither paper reports the other's metric, so the gap does not even have a determined sign.

Matched comparisons survive. The audit does not flag everything. CRAG's self-corrective variant is compared with Self-RAG at the same backbone, retriever, dataset and metric, and the 6.9-point gap stands. AstuteRAG evaluates RobustRAG's keyword-aggregation variant inside its own harness with one model and one retrieval setup, and the 4.4-point gap stands as an accuracy comparison, with the caveat that RobustRAG's certification guarantee is not what is being measured. Two of the three within-harness controls likewise pass unchanged. A comparison lens that rejected these would be useless.

Scope. These are cases, not a census. They were selected for the availability of both sides in local full text and for the presence of an explicit number, which favors open-access and widely cited work. They establish that the failure mode occurs, that it is decisive, and that it is sometimes repairable from the sources themselves; they do not estimate how often it occurs in the literature.

5 What Published Human Evaluations Report

Deciding cases like these is a judgment task, so we asked what would be needed to study it. Of 39 venue- or proceedings-linked Map B papers with locally extracted full text, a deterministic human-judgment term screen flagged 19, and full-text eligibility review retained nine that collected original human judgments. The released ledger records the exact text lines supporting every coding decision.

Two of nine papers report the number of distinct annotators. Two report both raters per item and the number of items ($R = 2$ and $P = 2$ in Appendix Table 12), four report an agreement or rater-uncertainty quantity, and two report compensation. None reports the item–annotator assignment, none releases pre-aggregation votes, and none randomizes an audit prompt across annotators, so this audit finds no verifiable assignment-unit mismatch. Absent fields are coded as reporting gaps, not as evidence that a practice was absent. The screen is narrow and single-coded and establishes concrete cases, not prevalence; the under-reporting finding itself is already established at scale by Howcroft et al. (2020) and van der Lee et al. (2021).

The specific increment is that the missing fields are exactly the ones a reanalysis needs. FaithEval reports three crowd workers per pair with majority inclusion (Ming et al., 2025); RAGTruth reports two annotators per response, a third reviewer for large differences, and $25 per hour (Niu et al., 2024); ParamMute reports six senior researchers and a multiply annotated 100-item subset but not their allocation to items (Huang et al., 2025); PoisonedRAG names the authors as evaluators without reviewer-level multiplicity (Zou et al., 2025). In every case the released artifact is a final aggregated label, enough to report an agreement rate and not enough to recompute uncertainty at the unit that was assigned. §7 prices that difference.

5.1 Development diagnostic and its negative result

We also asked whether the coordinates can be used directly as a coding instrument. In a development diagnostic, three coders reviewed 80 comparison claims and a fourth adjudicated them with model labels hidden. Coder agreement is uneven across coordinates: the policy/utility field is the most reliable (pairwise κ up to .737, Gwet's AC_1 up to .913) and the evidence state the least (κ .101–.283), with control status similarly low. Only 39/80 verdicts were unanimous.

Appendix Table 4 reports what those figures are worth against the baselines a classifier must beat. The full rule reaches .500 exact agreement, below the .537 obtained by predicting the modal adjudicated verdict, and adding the C2 and C5 coordinates to the ablation shifts exact agreement by .000 (95% CI $[-.066, +.079]$, exact McNemar $p = 1.0$); the five-coordinate variant reaches .550, which is also inside the majority-class interval. The sample overlaps codebook development and has a recoverable selection rule for only 16 items. We therefore state the negative result plainly: this diagnostic gives no evidence that the coordinates work as a four-way classifier, and the substantive evidence in this paper is the source-linked case

analysis of §4, not an instrument claim.

A later 480-claim protocol is retained only as a design case. One returned batch contained shared response payloads, and a later export could not be linked to controlled recruitment records, one-to-one pseudonym mappings, and separate response receipts available to the research team or an authorized auditor. We therefore exclude every response, contrast, reference statistic, participant attribute, and timing claim derived from it. Public identity disclosure is neither required nor desirable; the missing element was an internal audit trail, not public identifiers. The Ethics Statement gives the disclosure in full.

6 A Controlled Consequence Test

The two least conventional coordinates are origin/capability (C2) and policy/utility (C5), and the case analysis shows both doing real work. We therefore ask whether naming them changes a deployment decision, using a target that no human supplies.

Benchmark. A deterministic benchmark contains 600 synthetic deployment decisions in 300 counterfactual pairs, in three strata. In *C2* (120 pairs) only whether the deployed origin, access, and budget fall inside an evaluated capability envelope changes; half the pairs move both systems out of their envelopes and half move only one, so a capability shift can flip the ordering between systems rather than always resolving into deferral. In *C5* (120 pairs) only the error-versus-abstention prices or the exogenous price of commissioning a further study changes; the deferral price is displayed as a constant rather than derived from the deployment losses, so deferring is optimal on 65 of 240 items and the stratum is a genuine three-way decision. A *competence probe* (60 pairs) displays all three action losses directly and asks for the minimum. The target is always the unique minimum of the displayed loss rule, with a margin of at least .02; targets are balanced across the three actions within every stratum, so no stratum is solvable by a constant response.

Design. Each model answers every item under three arms crossed with four hand-written paraphrases, i.e. 7,200 greedy decisions per model. The *ablated* arm asks for evidence state, intervention, outcome, controls, and action. The *full* arm adds explicit origin/capability and policy/utility checks. The *placebo* arm adds generic reading instructions of matched length that name neither coordinate, so that *full* minus *placebo*—our primary contrast—is not confounded with instruction verbosity. Paraphrases vary wording, ordering, voice and layout. Uncertainty is a bootstrap clustered on the 300 counterfactual pairs, which is the unit the benchmark varies; the grid of gate-passing models by stratum receives a Holm correction, and models are combined by a random-effects meta-analysis so the reported effect refers to a model population.

Model	Probe	Gate	C2 Δ [95% CI]	C5 Δ [95% CI]
Qwen2.5-14B	95	yes	+4.8 [+3.3, +6.3]	+1.6 [+0.5, +2.7]
Qwen2.5-7B	89	yes	+11.1 [+8.8, +13.5]	-0.1 [-1.2, +1.0]
Qwen3-14B	97	yes	+14.3 [+11.6, +16.9]	+1.6 [+0.0, +3.1]
Qwen3-8B	70	yes	+6.0 [+4.3, +7.8]	+8.2 [+6.4, +10.2]
OLMo-2-13B	71	yes	+4.8 [+2.6, +7.1]	-3.1 [-4.6, -1.6]
OLMo-2-7B	45	no	-2.6 [-3.6, -1.6]	+7.8 [+6.7, +8.9]
Granite-3.1-8B	82	yes	+1.2 [+0.2, +2.3]	-0.6 [-1.3, +0.0]
Phi-4-mini	70	yes	+0.7 [-2.0, +3.5]	+2.1 [+0.3, +4.0]
Falcon3-7B	67	yes	+0.6 [-0.4, +1.7]	+2.0 [-0.1, +4.3]
Pooled (gate-passing)			+5.4 [+2.5, +8.2]	+1.4 [-0.4, +3.2]

Table 2: Controlled consequence test, version 2. Each model answers 600 items under three arms crossed with four paraphrases (7,200 greedy decisions per model). *Probe* is percentage accuracy on the displayed-loss competence stratum, and *Gate* records the pre-declared validity rule (probe $\geq 50\%$ in one arm and no single action on more than 90% of items). Δ is the primary contrast, *full* minus the length-matched *placebo* arm, in percentage points, with a percentile bootstrap interval clustered on the 300 counterfactual pairs. The pooled row is a DerSimonian–Laird random-effects combination over gate-passing models, so it refers to a model population rather than to any checkpoint. Appendix Table 18 adds the *full* minus *ablated* contrast, per-paraphrase spread, mean regret, and Holm-adjusted values.

Validity gate. A model enters the primary analysis only if it reaches 50% on the competence probe in at least one arm and does not emit a single action on more than 90% of items. This rule was fixed before the runs. It exists because a constant predictor is not performing the task, and counting one as evidence about prompt sensitivity would be a category error.

Results. 9 instruction-tuned open models from five developers were run; 8 passed the validity gate and 1 did not, with competence-probe accuracy ranging from 44.8% to 97.3%. Across gate-passing models the pooled *full* minus *placebo* effect is +5.4 points on the capability stratum (95% CI [+2.5, +8.2], $p < .001$, $I^2 = 96\%$) and +1.4 points on the utility stratum (95% CI [-0.4, +3.2], $p = .137$).

Appendix Table 18 gives the per-model contrasts, paraphrase spread, and Holm-adjusted values.

The competence probe is the internal negative control. Its items are decided by comparing three displayed losses, so naming a coordinate should not help, and it does not: the pooled effect there is +0.2 points (95% CI [-0.8, +1.2], $p = .682$). An arm effect concentrated on the strata where the coordinate is decisive, and absent where it is not, is what construct evidence for a coordinate should look like.

Two design points follow. The placebo arm matters: comparing *full* against *ablated* conflates coordinate naming with roughly eighty extra tokens of instruction, and the two contrasts do not agree. And between-model heterogeneity is large (I^2 above 90% on both substantive strata), which is exactly why a single-checkpoint result would have been uninformative. These are executable construct evidence about what naming a coordinate does to a model's deployment choice; they are not evidence about human auditors and do not validate the coordinates as a coding instrument.

7 Pricing the Reporting Gap

7.1 How heterogeneous are reviewers?

The consequence of analyzing at the wrong unit depends on how much reviewers differ. That quantity, σ_r, is the between-reviewer standard deviation of ability on the logit scale, and it is identified only by a design that crosses reviewers with items. We estimate it from the one such design available to us, the retained 80-claim diagnostic, whose three coders each labelled all 80 claims against an independent adjudication. A moment estimator that removes the binomial sampling component gives $\hat{\sigma}_r = 0.27$, with a source-clustered bootstrap 50% interval of $[0.23, 0.40]$ and 95% interval of $[0.00, 0.53]$. Three raters make a variance imprecise, so the interval is the estimate.

We then asked whether the published literature could do better. Two audited papers report a rater-agreement rate: RAGTruth reports 91.8% response-level consistency between two annotators, and FaithEval reports over 98% agreement under majority voting. Under the same latent model, a reported pairwise agreement constrains the total latent dispersion but not its split between reviewer and item components. At the label prevalence those papers imply, every σ_r on a 0–1.2 grid reproduces the reported agreement for some item dispersion: the reported statistic places no usable restriction at all (Appendix Table 14). Separating the components requires the item–annotator assignment, which no audited paper reports. The sweep below is therefore not a stylistic choice; it is what the available reporting permits.

7.2 Design

We simulate a two-arm claim-review study with twelve reviewers, six per prompt, G sources contributing six claims each, and a balanced three-of-six panel per claim in both arms. Baseline agreement is 75%; source and claim logit standard deviations are .15 and .35. Reviewer abilities are redrawn from $N(0, \sigma_r^2)$ in every replicate, so the target is a reviewer-population effect and the source-only analysis cannot be defended as answering a realized-panel estimand instead; Appendix Table 15 repeats everything with abilities held at fixed population quantiles, with the same conclusions. Each cell uses 5,000 replicates at $\alpha = .05$, under a sharp null and under a true prompt effect of 0.4 logits.

Four analyses are compared. *Panel/src* is the status quo: a paired source t test on panel-majority correctness, computable from released majority labels alone. *Votes/src* keeps the source-level standard error but uses individual votes. *Rev. t* is a two-sample t test on the twelve individual reviewer accuracies, and *Rev. RI* refers that contrast to all $\binom{12}{6} = 924$ arm assignments. The last two need only pre-aggregation votes carrying reviewer identifiers, so unlike an oracle test over a complete vote matrix they are executable after collection.

7.3 Results

At the estimated $\hat{\sigma}_r = 0.27$ and 80 sources, the status-quo analysis rejects a true null 24.2% of the time, and 36.7% at 160 sources: adding sources makes it worse, because conditional item precision improves while the number of prompt assignments stays at twelve. Replacing majority labels with individual votes does not help and in fact hurts, reaching 32.9% at 80 sources. That comparison isolates the cause. The defect is the unit over which uncertainty is computed, not the majority aggregation, which is why advice to release pre-adjudication labels is necessary but not sufficient.

Both reviewer-level analyses hold the nominal level across the whole grid: 4.6% and 4.3% at $\hat{\sigma}_r = 0.27$ and 80 sources, and 4.4% and 4.3% at 160 sources, with the same behavior at σ_r up to

Setting	Panel/src	Votes/src	Rev. t	Rev. RI
Size: no prompt effect, $\alpha = .05$				
$\sigma_r = 0.00, G = 80$	4.7	5.1	4.7	4.3
$\sigma_r = 0.23, G = 80$	20.6	27.9	4.9	4.7
$\sigma_r = 0.27, G = 80$ *	24.2	32.9	4.6	4.3
$\sigma_r = 0.40, G = 80$	38.6	49.0	5.7	5.4
$\sigma_r = 0.53, G = 80$	48.8	57.3	5.1	5.1
$\sigma_r = 0.27, G = 20$	9.9	13.3	4.8	3.7
$\sigma_r = 0.27, G = 40$	14.7	21.5	5.0	4.3
$\sigma_r = 0.27, G = 160$	36.7	46.5	4.4	4.3
Power: true prompt effect 0.4 *logits*				
$\sigma_r = 0.27, G = 80$	78.8	89.0	52.8	51.6
$\sigma_r = 0.27, G = 160$	89.8	94.9	56.3	55.4

Table 3: Rejection percentages over 5,000 replicates with twelve reviewers, six per prompt, six claims per source G, and balanced three-of-six panels. Reviewer abilities are redrawn every replicate, so the target is a reviewer-population effect rather than a realized-panel contrast. σ_r is the between-reviewer ability standard deviation on the logit scale; the starred row is the value estimated from the crossed three-coder diagnostic ($\hat{\sigma}_r = 0.27$, Appendix Table 14). *Panel/src* is the status-quo paired source t test on panel-majority correctness; *Votes/src* keeps the source-level standard error but uses individual votes; *Rev. t* is a two-sample t test on the twelve individual reviewer accuracies; *Rev. RI* refers the same contrast to all 924 arm assignments. The last two are executable retrospectively from released pre-aggregation votes with reviewer identifiers, and retain power.

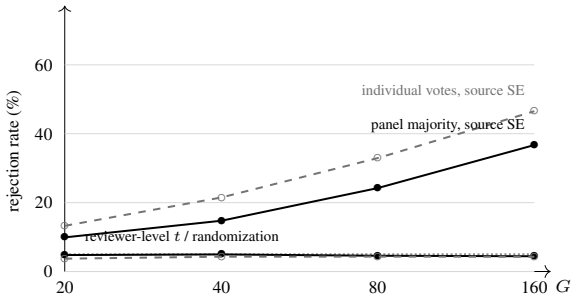


Figure 2: Null rejection rate against the number of sources G at the estimated reviewer heterogeneity $\sigma_r = 0.27$, with reviewer abilities redrawn each replicate. Adding sources makes the two source-level analyses steadily worse, and using individual votes instead of majority labels does not help: the defect is the unit of uncertainty, not the aggregation. The two reviewer-level analyses stay at the nominal level and are executable from released pre-aggregation votes. Table 3 gives the values and the corresponding power.

0.53 (Figure 2). They are not merely conservative: against a true prompt effect of 0.4 logits they reject 52.8% and 51.6% at 80 sources, rising to 56.3% and 55.4% at 160. The valid analyses lose power relative to the invalid ones, but that difference is the part of the apparent precision that was never earned.

What an author of a collected study should do. Compute the effect at the reviewer level and report it two ways: a two-sample test on individual reviewer accuracies, and its exact assignment permutation. Both require only that pre-aggregation votes be released with stable reviewer identifiers. Where reviewers are crossed with items, a mixed-effects model with crossed reviewer and item intercepts targets the same estimand (Baayen et al., 2008; Barr et al., 2013); our contribution is not that recommendation but the demonstration that the simpler reviewer-level tests already recover calibration in this design, and that the single artifact blocking either of them is the one no audited paper releases.

8 A Pre-Collection Checklist

The audit is cheapest before labels exist. Five decisions follow from §4–§7; Appendix Table 26 expands the reporting fields. *(1) Name the claim and decision.* The unit under review is an atomic public statement, not a paper, and verifying source support, deciding comparability, and recommending an action are three different targets. *(2) Draw the unit map.* List the item, reviewer, treatment-assignment, scoring, reference, and generalization units together; if prompts are assigned to people, people remain the treatment units after majority voting, and claim-in-paper nesting adds a layer rather than replacing reviewer dependence. *(3) Construct the target independently.* A reference rubric must not paraphrase the tested intervention; where possible generate the target from a predeclared consequence such as deployment loss under matched capability and cost assumptions, and prefer admissible sets to forced adjudication when the sources are ambiguous. *(4) Power the randomized units.* Vary the number and heterogeneity of reviewers, not only the number of claims: if an interval keeps shrinking while reviewers are held fixed, it is conditional on the panel (Card et al., 2020). Our $\hat{\sigma}_r = 0.27$ is a usable starting value. *(5) Preserve the measurement path.* Release pre-adjudication votes with stable reviewer identifiers, class supports, abstentions, unresolved cases, and aggregation rules. §7 shows that this single artifact is what makes a valid retrospective analysis possible; majority labels alone do not. Hashes establish

which bytes were analyzed, not that nominally distinct people acted independently, and unverifiable provenance requires exclusion rather than statistical repair.

A prospective human test. A corrected experiment would randomize enough independent participants between prompts, or cross participants with prompts under counterbalancing and carryover controls, analyze at the assignment unit, and model participant, item, and reference-rater variation, with the protocol externally timestamped before outcomes (van Miltenburg et al., 2021). The benchmark of §6 supplies an intervention-independent target; §7 supplies the analysis plan and a starting value for the power calculation.

9 Conclusion

Robust RAG is not a single leaderboard, and the cost of pretending otherwise is measurable. Ten of twelve cross-paper comparisons drawn from published robust-RAG work do not survive a five-coordinate audit as stated, and in the sharpest case a quoted 3.7-point attack-success lead reverses to -45.3 points once both sides are read at a matched insertion budget. Two of three within-harness contrasts pass unchanged, so the coordinates discriminate rather than condemn. Matching capability envelopes and outcome units is not a methodological nicety; it decides orderings.

Deciding such claims with human reviewers is itself a measurement problem, and the field currently cannot check its own answers. None of the nine robust-RAG papers that collect original human judgments reports the item–annotator assignment or releases pre-aggregation votes. That is not a bookkeeping complaint: at the reviewer heterogeneity we estimate, the analysis those papers make possible rejects a true null roughly a quarter of the time at 80 sources and gets worse as items accumulate, while two reviewer-level analyses hold the nominal level and keep power. Both are executable after collection and both need exactly the artifact that is missing. Releasing per-reviewer votes is therefore the single highest-value change available, and it costs nothing.

We are equally clear about what is not shown: the coordinates are disclosure fields validated on published comparisons, not a coding instrument, since on an 80-claim development sample no variant of the schema separates from predicting the modal verdict.

Artifact availability. The supplement releases the source-linked comparability cases and their verifier, the published-study ledger, the v2 benchmark, model outputs and clustered analysis, the reviewer-heterogeneity estimator, and both simulations. Every evidence-bearing table is linked to a machine-readable source. Excluded outcome exports are omitted from release packages; a privacy-safe invalidation receipt preserves the shared-payload finding and the original return-manifest hash.

Limitations

The twelve comparability cases are cases, not a census. Both sides had to be present in local full text with an explicit number, which favors open-access and widely cited work, and the coordinate-matched re-reading is available only where a paper reports an operating point at the other side's setting; seven of the fifteen entries record that no such value is recoverable from the text. Coding of the cases is ours and single-coded, though every quoted value is machine-verified against its source line. The cases show that the failure mode occurs and can be decisive; they do not estimate its frequency.

The nine-paper reporting audit is a single-coded, model-assisted pass over 39 venue-linked local full texts. It is neither a scoping review conducted to protocol (Tricco et al., 2018) nor a risk-of-bias assessment (Whiting et al., 2011), and the under-reporting phenomenon is established at larger scale elsewhere (Howcroft et al., 2020; van der Lee et al., 2021). Absence of a reported field is not evidence of an absent practice, and later supplements or author records could resolve fields marked unreported. The broader map is open-PDF based and overrepresents English-language, openly accessible work.

The reviewer-heterogeneity estimate rests on three coders and 80 claims: the bootstrap interval $[0.00, 0.53]$ is wide, and the point value 0.27 should be read as a starting value, not a field constant. The simulation uses binary correctness, a common wrong-label state, balanced three-of-six panels, and a logit random-effects model; it demonstrates a design failure mode and the analyses that repair it, not a prevalence. The two reviewer-level remedies require released per-reviewer votes; where reviewers rate disjoint item sets they will be less efficient than a crossed mixed-effects model.

The controlled benchmark is templated arithmetic prose with a synthetic loss rule. Open instruction-tuned checkpoints are a convenience

sample of a model population, results do not predict behavior on natural paper claims, and the paraphrase set is four hand-written variants rather than a sampled template population. The v1 benchmark was corrected after outcomes were observed (a C2 generator also moved the deferral cost); v2 was rebuilt and every reported model was rerun on it, so the v2 contrast is a fresh measurement but the benchmark family is not preregistered.

The 80-claim diagnostic is purposive, overlaps codebook development, and lacks a recoverable selection procedure for 64 items; its result is negative and is reported as such. The excluded 480-claim collection lacks the internal provenance needed for a human-study claim, and no quantity derived from it appears anywhere in this paper, including the heterogeneity estimate. Local hashes authenticate bytes, not behavioral independence or external preregistration. No consequence-based human effect size is estimated.

Ethics Statement

For the retained 80-claim diagnostic, the project owner attests that all four contributors consented to paper use and public release of de-identified annotations, including rationales, were compensated, and did not use generative AI for annotation. The public provenance is that attestation rather than contributor-level consent or identity records. The project owner also reports an institutional determination that formal review was not required.

We make no human-subject, recruitment, expertise, compensation, or independence claim for the excluded 480-claim collection. One returned batch contained shared response payloads, and a later export could not be linked to controlled recruitment records, one-to-one pseudonym mappings, and separate response receipts. Because that internal audit trail was absent, every response, contrast, reference statistic, participant attribute, and timing claim derived from it is excluded, including from the reviewer-heterogeneity estimate in §7; a privacy-safe invalidation receipt preserves the shared-payload finding and the original return-manifest hash. We distinguish public de-identification from private auditability: consent, recruitment, and identity-linkage records may remain access-controlled, but an empirical claim still needs an inspectable internal provenance path. The published-paper audit uses public scholarly documents and introduces no participant contact. The comparability cases quote short passages from published papers for verification purposes only. LLMs assisted documentary discovery, extraction, and audit screening; those outputs are not human gold. The synthetic benchmark contains no participant data or third-party quotation. The reported open-model runs use deterministic decoding and release raw outputs. No attack payloads or poisoned corpora are released.

References

Alberto Abadie, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge. 2023. When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35.

Klejda Alushi, Jan Strich, Chris Biemann, and Martin Semmann. 2026. Comprehensive comparison of RAG methods across multi-domain conversational QA. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 233–247, Rabat, Morocco. Association for Computational Linguistics.

Ron Artstein and Massimo Poesio. 2008. Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *ICLR*.

R. Harald Baayen, Douglas J. Davidson, and Douglas M. Bates. 2008. Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language*, 59(4):390–412.

Gaurav Bagwe, Saket S. Chaturvedi, Xiaolong Ma, Xiaoyong Yuan, Kuang-Ching Wang, and Lan Zhang. 2025. Your rag is unfair: Exposing fairness vulnerabilities in retrieval-augmented generation via backdoor attacks.

Dale J. Barr, Roger Levy, Christoph Scheepers, and Harry J. Tily. 2013. Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68(3):255–278.

Anya Belz, Shubham Agarwal, Anastasia Shimorina, and Ehud Reiter. 2021. A systematic review of reproducibility research in natural language processing. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 381–393.

Anya Belz, Craig Thomson, Ehud Reiter, Gavin Abercrombie, Jose M. Alonso-Moral, Mohammad Arvan, and 1 others. 2023. Missing information, unresponsive authors, experimental flaws: The impossibility of assessing the reproducibility of previous human evaluations in NLP. In *Proceedings of the Fourth Workshop on Insights from Negative Results in NLP*, pages 1–10.

Emily M. Bender and Batya Friedman. 2018. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604.

Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego de Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, and 9 others. 2022. Improving language models by retrieving from trillions of tokens. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 2206–2240. PMLR.

Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. 2020. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 9263–9274.

Chia-Yuan Chang, Zhimeng Jiang, Vineeth Rakesh, Menghai Pan, Chin-Chia Michael Yeh, Guanchu Wang, Mingzhi Hu, Zhichao Xu, Yan Zheng, Mahashweta Das, and Na Zou. 2025. MAIN-RAG: Multi-agent filtering retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2607–2622. Association for Computational Linguistics.

Hongyan Chang, Ergute Bao, Xinjian Luo, and Ting Yu. 2026. Overcoming the retrieval barrier: Indirect prompt injection in the wild for llm systems.

Saket S. Chaturvedi, Gaurav Bagwe, Lan Zhang, and Xiaoyong Yuan. 2025. Aip: Subverting retrieval-augmented generation via adversarial instructional prompt.

Harsh Chaudhari, Giorgio Severi, John Abascal, Anshuman Suri, Matthew Jagielski, Christopher A. Choquette-Choo, Milad Nasr, Cristina Nita-Rotaru, and Alina Oprea. 2024. Phantom: General backdoor attacks on retrieval augmented language generation.

Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. 2017. Reading Wikipedia to answer open-domain questions. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1870–1879. Association for Computational Linguistics.

Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. 2024a. Benchmarking large language models in retrieval-augmented generation. In *AAAI*.

Liuji Chen, Xiaofang Yang, Yuanzhuo Lu, Jinghao Zhang, Xin Sun, Qiang Liu, Shu Wu, Jing Dong, and Liang Wang. 2025a. Poisonarena: Uncovering competing poisoning attacks in retrieval-augmented generation.

Lu Chen, Ruqing Zhang, Jiafeng Guo, Yixing Fan, and Xueqi Cheng. 2024b. Controlling risk of retrieval-augmented generation: A counterfactual prompting framework. In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

Zhuo Chen, Yuyang Gong, Jiawei Liu, Miaokun Chen, Haotan Liu, Qikai Cheng, Fan Zhang, Wei Lu, and Xiaozhong Liu. 2025b. Flippedrag: Black-box opinion manipulation adversarial attacks to retrieval-augmented generation models.

Pengzhou Cheng, Yidong Ding, Tianjie Ju, Zongru Wu, Wei Du, Ping Yi, Zhuosheng Zhang, and Gongshen Liu. 2024. Trojanrag: Retrieval-augmented generation can be backdoor driver in large language models.

Sarthak Choudhary, Nils Palumbo, Ashish Hooda, Krishnamurthy Dj Dvijotham, and Somesh Jha. 2025. Through the stealth lens: Attention-aware defenses against poisoning in rag.

C. K. Chow. 1970. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46.

Elizabeth Clark, Tal August, Sofia Serrano, Nikita Haduong, Suchin Gururangan, and Noah A. Smith. 2021. All that's 'human' is not gold: Evaluating human evaluation of generated text. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7282–7296.

Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. 2022. Dealing with disagreements: Looking beyond the majority vote in subjective annotations. *Transactions of the Association for Computational Linguistics*, 10:92–110.

A. P. Dawid and A. M. Skene. 1979. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(1):20–28.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 11733–11763. PMLR.

Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. 2024. RAGAs: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158.

Feiteng Fang, Yuelin Bai, Shiwen Ni, Min Yang, Xiaojun Chen, and Ruifeng Xu. 2024. Enhancing noise robustness of retrieval-augmented language models with adaptive adversarial training. In *ACL*.

Jorge Gabín, Anxo Perez, and Javier Parapar. 2026. Lost in the evidence? reproducing document position and context size effects in rag.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488. Association for Computational Linguistics.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. Retrieval-augmented generation for large language models: A survey.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *ICML*.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. REALM: Retrieval-augmented language model pre-training. In *ICML*.

David M. Howcroft, Anya Belz, Miruna-Adriana Clinciu, Dimitra Gkatzia, Sadid A. Hasan, Saad Mahamood, Simon Mille, Emiel van Miltenburg, Sashank Santhanam, and Verena Rieser. 2020. Twenty years of confusion in human evaluation: NLG needs evaluation sheets and standardised definitions. In *Proceedings of the 13th International Conference on Natural Language Generation*, pages 169–182.

Wentao Hu, Wengyu Zhang, Yiyang Jiang, Chen Jason Zhang, Xiaoyong Wei, and Li Qing. 2025. Removal of hallucination on hallucination: Debate-augmented RAG. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15839–15853. Association for Computational Linguistics.

Pengcheng Huang, Zhenghao Liu, Yukun Yan, Haiyan Zhao, Xiaoyuan Yi, Hao Chen, Zhiyuan Liu, Maosong Sun, Tong Xiao, Ge Yu, and Chenyan Xiong. 2025. Parammute: Suppressing knowledge-critical ffns for faithful retrieval-augmented generation. In *Advances in Neural Information Processing Systems*.

Yukun Huang, Sanxing Chen, Hongyi Cai, and Bhuwan Dhingra. 2024. To trust or not to trust? enhancing large language models' situated faithfulness to external contexts.

Jeongyeon Hwang, Junyoung Park, Hyejin Park, Dongwoo Kim, Sangdon Park, and Jungseul Ok. 2025. Retrieval-augmented generation with estimation of source reliability. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 34279–34303.

Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *EACL*.

Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. 2023. Atlas: Few-shot learning with retrieval augmented language models. *Journal of Machine Learning Research*, 24(251):1–43.

Chaeyun Jang, Deukhwan Cho, Seanie Lee, Hyungi Lee, and Juho Lee. 2025. Reliable decision-making via calibration-oriented retrieval-augmented generation. In *Advances in Neural Information Processing Systems*.

Bowen Jin, Jinsung Yoon, Jiawei Han, and Sercan O. Arik. 2024a. Long-context llms meet rag: Overcoming challenges for long inputs in rag.

Zhuoran Jin, Pengfei Cao, Yubo Chen, Kang Liu, Xiaojian Jiang, Jiexin Xu, Qiuxia Li, and Jun Zhao. 2024b. Tug-of-war between knowledge: Exploring and resolving knowledge conflicts in retrieval-augmented language models.

Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. 2025. Sufficient context: A new lens on retrieval augmented generation systems. In *International Conference on Learning Representations*.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *EMNLP*.

Jungo Kasai, Keisuke Sakaguchi, Yoichi Takahashi, Ronan Le Bras, Akari Asai, Xinyan Yu, Dragomir Radev, Noah Smith, Yejin Choi, and Kentaro Inui. 2023. RealTime QA: What's the answer right now? In *Advances in Neural Information Processing Systems*, volume 36, pages 49025–49043.

Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. 2020. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations*.

Tiziano Labruna, Jon Ander Campos, and Gorka Azkune. 2024. When to retrieve: Teaching llms to utilize information retrieval effectively.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *NeurIPS*.

Shuowei Li, Haoxin Li, Wenda Chu, and Yi Fang. 2026. How large language models balance internal knowledge with user and document assertions.

Xun Liang, Simin Niu, Zhiyu Li, Sensen Zhang, Hanyu Wang, Feiyu Xiong, Jason Zhaoxin Fan, Bo Tang, Shichao Song, Mengwei Wang, and Jiawei Yang. 2025. Saferag: Benchmarking security in retrieval-augmented generation of large language model.

Gabrielle Kaili-May Liu, Bryan Li, Arman Cohan, William Gantt Walden, and Eugene Yang. 2025a. Investigating retrieval-augmented generation systems on unanswerable, uncheatable, realistic, multi-hop queries.

Huanshuo Liu, Hao Zhang, Zhijiang Guo, Jing Wang, Kuicai Dong, Xiangyang Li, Yi Quan Lee, Cong Zhang, and Yong Liu. 2025b. Ctrla: Adaptive retrieval-augmented generation via inherent control. In *Findings of ACL*.

Nelson Liu, Tianyi Zhang, and Percy Liang. 2023. Evaluating verifiability in generative search engines. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 7001–7025. Association for Computational Linguistics.

Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. 2021. Entity-based knowledge conflicts in question answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7052–7063. Association for Computational Linguistics.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822. Association for Computational Linguistics.

Yifei Ming, Senthil Purushwalkam, Shrey Pandit, Zixuan Ke, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. 2025. Faitheval: Can your language model stay faithful to context, even if "the moon is made of marshmallows". In *International Conference on Learning Representations*.

Vinh Nguyen, Cuong Dang, Jiahao Zhang, Hoa Tran, Minh Tran, Trinh Chau, Thai Le, Lu Cheng, and Suhang Wang. 2026. Urag: A benchmark for uncertainty quantification in retrieval-augmented large language models.

Bo Ni, Zheyuan Liu, Leyao Wang, Yongjia Lei, Yuying Zhao, Xueqi Cheng, Qingkai Zeng, Luna Dong, Yinglong Xia, Krishnaram Kenthapadi, Ryan Rossi, Franck Dernoncourt, Md Mehrab Tanjim, Nesreen Ahmed, Xiaorui Liu, Wenqi Fan, Erik Blasch, Yu Wang, Meng Jiang, and Tyler Derr. 2025. Towards trustworthy retrieval augmented generation for large language models: A survey.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, Kashun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.

Nicolas Papernot, Patrick McDaniel, Arunesh Sinha, and Michael P. Wellman. 2018. SoK: Security and privacy in machine learning. In *2018 IEEE European Symposium on Security and Privacy*, pages 399–414. IEEE.

Rebecca J. Passonneau and Bob Carpenter. 2014. The benefits of a model of annotation. *Transactions of the Association for Computational Linguistics*, 2:311–326.

Silviu Paun, Bob Carpenter, Jon Chamberlain, Dirk Hovy, Udo Kruschwitz, and Massimo Poesio. 2018. Comparing bayesian models of annotation. *Transactions of the Association for Computational Linguistics*, 6:571–585.

Sezen Percin, Xin Su, Qutub Sha Syed, Phillip Howard, Aleksei Kuvshinov, Leo Schwinn, and Kay-Ulrich Scholl. 2025. Investigating the robustness of retrieval-augmented generation at the query level. In *GEM*.

Fabio Petroni, Aleksandra Piktus, Angela Fan, Patrick Lewis, Majid Yazdani, Nicola De Cao, James Thorne, Yacine Jernite, Vladimir Karpukhin, Jean Maillard, Vassilis Plachouras, Tim Rocktäschel, and Sebastian Riedel. 2021. KILT: a benchmark for knowledge intensive language tasks. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2523–2544. Association for Computational Linguistics.

Barbara Plank. 2022. The "problem" of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 10671–10682.

W. Scott Richardson, Mark C. Wilson, Jim Nishikawa, and Robert S. A. Hayward. 1995. The well-built clinical question: A key to evidence-based decisions. *ACP Journal Club*, 123(3):A12.

Sara Rosenthal, Yannis Katsis, Vraj Shah, Lihong He, Lucian Popa, and Marina Danilevsky. 2026. Mtrag-un: A benchmark for open challenges in multi-turn rag conversations.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2024. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 338–354.

Jakob Schuster, Vagrant Gautam, and Katja Markert. 2026. Whose facts win? llm source preferences under knowledge conflicts.

Chaitanya Sharma. 2025. Retrieval-augmented generation: A comprehensive survey of architectures, enhancements, and robustness frontiers.

Weijia Shi, Sewon Min, Michihiro Yasunaga, Minjoon Seo, Richard James, Mike Lewis, Luke Zettlemoyer, and Wen-tau Yih. 2024. REPLUG: Retrieval-augmented black-box language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8371–8384. Association for Computational Linguistics.

Xiaonan Si, Meilin Zhu, Simeng Qin, Lijia Yu, Lijun Zhang, Shuaitong Liu, Xinfeng Li, Ranjie Duan, Yang Liu, and Xiaojun Jia. 2025. Secon-rag: A two-stage semantic filtering and conflict-free framework for trustworthy rag. In *Advances in Neural Information Processing Systems*.

Hongru Song, Yu-An Liu, Ruqing Zhang, Jiafeng Guo, Jianming Lv, Maarten de Rijke, and Xueqi Cheng. 2025. The silent saboteur: Imperceptible adversarial attacks against black-box retrieval-augmented generation systems.

Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. 2024. ConflictBank: A benchmark for evaluating the influence of knowledge conflicts in LLMs. In *Advances in Neural Information Processing Systems*, volume 37, pages 103242–103268.

Giovanni Trappolini, Florin Cuconasu, Simone Filice, Yoelle Maarek, and Fabrizio Silvestri. 2025. Redefining retrieval evaluation in the era of llms.

Andrea C. Tricco, Erin Lillie, Wasifa Zarin, Kelly K. O'Brien, Heather Colquhoun, Danielle Levac, David Moher, Micah D. J. Peters, Tanya Horsley, Laura Weeks, Susanne Hempel, Elie A. Akl, Christine Chang, Jessie McGowan, Lesley Stewart, Lisa Hartling, Adrian Aldcroft, Michael G. Wilson, Chantelle Garritty, and 9 others. 2018. PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. *Annals of Internal Medicine*, 169(7):467–473.

Alexandra N. Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470.

Chris van der Lee, Albert Gatt, Emiel van Miltenburg, and Emiel Krahmer. 2021. Human evaluation of automatically generated text: Current trends and best practice guidelines. *Computer Speech & Language*, 67:101151.

Emiel van Miltenburg, Chris van der Lee, and Emiel Krahmer. 2021. Preregistering NLP research. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 613–623.

Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny

Zhou, Quoc Le, and Thang Luong. 2024. Fresh-LLMs: Refreshing large language models with search engine augmentation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 13697–13720. Association for Computational Linguistics.

Fei Wang, Xingchen Wan, Ruoxi Sun, Jiefeng Chen, and Sercan O. Arik. 2025a. Astute rag: Overcoming imperfect retrieval augmentation and knowledge conflicts for large language models. In *ACL*.

Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. 2025b. Retrieval-augmented generation with conflicting evidence. *Preprint*, arXiv:2504.13079.

Xiaohua Wang, Zhenghua Wang, Xuan Gao, Feiran Zhang, Yixin Wu, Zhibo Xu, Tianyuan Shi, Zhengyuan Wang, Shizheng Li, Qi Qian, Ruicheng Yin, Changze Lv, Xiaoqing Zheng, and Xuanjing Huang. 2024. Searching for best practices in retrieval-augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17716–17736. Association for Computational Linguistics.

Zhiruo Wang, Jun Araki, Zhengbao Jiang, Md Rizwan Parvez, and Graham Neubig. 2023. Learning to filter context for retrieval-augmented generation.

Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. 2025. Know your limits: A survey of abstention in large language models. *Transactions of the Association for Computational Linguistics*, 13:529–556.

Penny F. Whiting, Anne W. S. Rutjes, Marie E. Westwood, Susan Mallett, Jonathan J. Deeks, Johannes B. Reitsma, Mariska M. G. Leeflang, Jonathan A. C. Sterne, and Patrick M. M. Bossuyt. 2011. QUADAS-2: A revised tool for the quality assessment of diagnostic accuracy studies. *Annals of Internal Medicine*, 155(8):529–536.

Chong Xiang, Tong Wu, Zexuan Zhong, David Wagner, Danqi Chen, and Prateek Mittal. 2024. Certifiably robust rag against retrieval corruption.

Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8541–8565, Miami, Florida, USA. Association for Computational Linguistics.

Yuming Xu, Mingtao Zhang, Zhuohan Ge, Haoyang Li, Nicole Hu, Yongqi Zhang, Zhiyuan Wen, Jason Chen Zhang, Qing Li, and Lei Chen. 2026. Securing retrieval-augmented generation: A taxonomy of attacks, defenses, and future directions.

Jiaqi Xue, Mengxin Zheng, Yebowen Hu, Fei Liu, Xun Chen, and Qian Lou. 2024. Badrag: Identifying vulnerabilities in retrieval augmented generation of large language models.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. 2024. Corrective retrieval augmented generation.

Xiangyu Yin, Yi Qi, and Chih-Hong Cheng. 2026. Progrank: Probe-gradient reranking to defend dense-retriever rag from corpus poisoning.

Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2024. Making retrieval-augmented language models robust to irrelevant context. In *International Conference on Learning Representations*.

Hao Yu, Aoran Gan, Kai Zhang, Shiwei Tong, Qi Liu, and Zhaofeng Liu. 2024. Evaluation of retrieval-augmented generation: A survey.

Wenhao Yu, Hongming Zhang, Xiaoman Pan, Kaixin Ma, Hongwei Wang, and Dong Yu. 2023. Chain-of-note: Enhancing robustness in retrieval-augmented language models.

Baolei Zhang, Haoran Xin, Minghong Fang, Zhuqing Liu, Biao Yi, Tong Li, and Zheli Liu. 2025a. Traceback of poisoned texts in poisoning attacks to retrieval-augmented generation.

Qinggang Zhang, Zhishang Xiang, Yilin Xiao, Le Wang, Junhui Li, Xinrun Wang, and Jinsong Su. 2025b. Faithfulrag: Fact-level conflict modeling for context-faithful retrieval-augmented generation. In *ACL*.

Yuxin Zhang, Yan Wang, Yongrui Chen, Shenyu Zhang, Xinbang Dai, Sheng Bi, and Guilin Qi. 2025c. Magic mushroom: A customizable benchmark for fine-grained analysis of retrieval noise erosion in rag systems.

Zexuan Zhong, Ziqing Huang, Alexander Wettig, and Danqi Chen. 2023. Poisoning retrieval corpora by injecting adversarial passages.

Yujia Zhou, Wenbo Zhang, Jingying Shao, Yan Liu, Xiaoxi Li, Jiajie Jin, Hongjin Qian, Zheng Liu, Chaozhuo Li, Jason Chen Zhang, Zhicheng Dou, Philip S. Yu, and Jiaxin Mao. 2024. Trustworthiness in retrieval-augmented generation systems: A survey.

Xi Zhu, Ziqi Wang, Kai Mei, Wujiang Xu, Minghao Guo, Bangji Yang, Jiajun Fan, and Dimitris N Metaxas. 2026. Trust or abstain? a self-aware rag approach.

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. 2025. Poisonedrag: Knowledge corruption attacks to retrieval-augmented generation of large language models. In *USENIX Security*.

A Substantive Comparability in Robust RAG

We use the literature map as a source of boundary cases, not as prevalence evidence. Five *operating questions* organize discovery: how evidence is constructed, how a model uses it, how deliberate corruption changes the game, how signals become actions, and how those actions are evaluated. The *comparison coordinates* in Section 3 are different: they describe one proposed contrast. The five *measurement objects* used later—item, reviewer, assignment, reference, and decision—describe how that contrast is judged. Thus, the three sets of five operate at literature, claim, and measurement levels rather than naming competing taxonomies.

Evidence availability is not evidence use. Irrelevant-context studies test whether retrieved distractors redirect a generator (Yoran et al., 2024); noise-aware training changes the generator's training distribution (Fang et al., 2024); RGB supplies diagnostic stress conditions (Chen et al., 2024a); and Chain-of-Note changes how retrieved documents are read (Yu et al., 2023). A filter, a trained reader, and a benchmark may share a harness, but their native scores do not rank interchangeable interventions. Insufficient evidence creates another boundary: when no passage can support an answer, re-retrieval or abstention can dominate better distractor resistance (Joren et al., 2025).

Support, authority, conflict, and attack are distinct. RAGTruth annotates unsupported spans, while evaluators estimate claim or response support (Niu et al., 2024; Es et al., 2024; Saad-Falcon et al., 2024); knowledge-conflict methods instead study competition between parametric memory and retrieved evidence (Jin et al., 2024b; Zhang et al., 2025b; Wang et al., 2025a). A response can be correct but unsupported, or faithful to a weak source. Provenance and timestamp determine which source should govern (Petroni et al., 2021; Vu et al., 2024; Kasai et al., 2023). Deliberate corruption changes the game again: corpus poisoning, indirect prompt injection, and backdoors differ in corpus write access, query knowledge, retriever visibility, insertion budget, persistence, and objective (Zou et al., 2025; Chaturvedi et al., 2025; Chaudhari et al., 2024), and their defenses prevent, detect, certify, trace, quarantine, or recover under different assumptions (Xiang et al., 2024; Si et al., 2025; Zhang et al., 2025a). An attack-success comparison is transportable only within a matched capability envelope.

A signal is not a policy. Calibration predicts a property of a score (Guo et al., 2017); selective prediction trades coverage against risk (Geifman and El-Yaniv, 2017); sufficient-context classification asks whether available passages can support an answer; adaptive retrieval chooses whether to search, verify, answer, or abstain (Jang et al., 2025; Joren et al., 2025; Labruna et al., 2024). Self-RAG, CRAG, and CtrlA expose this policy layer explicitly (Asai et al., 2024; Yan et al., 2024; Liu et al., 2025b). Comparing them requires a common action set, budget, and loss: accuracy alone cannot price extra retrieval, coverage, or refusal.

Comparison ladder. The strongest contrast reruns alternatives with the same corpus snapshot, retriever, generator, evidence state, capability envelope, action set, and outcome. When that is impossible, synthesis should retain only the common scope supported by both studies. If a decision-critical field is absent, the comparison is underspecified rather than evidence that either method is better. Appendix B gives paper-level contrasts and the documentary-map boundary; the next section turns these recurring boundaries into a claim-level audit.

B Extended Survey by Scientific Question

This appendix expands the compact domain context in the main paper through representative paper-level contrasts. It is not an exhaustive bibliography; the released 342-row evidence map contains the long-tail records, source strata, and intervention labels. Because the map has no study-level risk-of-bias weighting and its initial search logs are incomplete, the synthesis below identifies recurring scientific distinctions rather than estimating prevalence or ranking evidence families. The review-directed recall check on 2026-07-15 also identified older, pre-freeze publications absent from the frozen map. We cite those papers as qualitative context and log them as post-freeze discoveries; they do not enter the 342-row denominator.

B.1 When Does Retrieved Context Help?

Irrelevant-context and noise-aware-training studies intervene at different points. Yoran et al. (2024) shows that retrieved distractors can actively redirect generation and tests filtering and targeted fine-tuning. Fang et al. (2024) instead trains a generator across random, relevant, and counterfactual

retrieval conditions. RGB supplies stress conditions and diagnostic abilities rather than a competing mitigation (Chen et al., 2024a). These papers can share an end-to-end harness, but their native results do not rank a filter, a trained reader, and a benchmark as interchangeable objects.

Context construction creates a second boundary. Chain-of-Note changes how the generator reads retrieved documents, whereas compression and reranking alter what reaches it (Yu et al., 2023; Wang et al., 2023; Trappolini et al., 2025). Long-context studies show that retaining more passages can improve evidence availability while worsening position sensitivity or distraction (Jin et al., 2024a; Gabín et al., 2026). A fair comparison therefore fixes the candidate evidence, generator, decoding, and budget when the intended claim is about the intervention rather than the entire system.

Insufficient evidence is not merely severe distraction. Sufficient Context labels whether the available passages can support an answer and evaluates selective behavior (Joren et al., 2025). A reader that resists a hard negative and a controller that abstains when support is absent solve different problems. Reports should state both answer-bearing support availability and the scored action.

B.2 Which Knowledge Should Govern an Answer?

Knowledge-conflict work separates context–memory disagreement from conflict among retrieved sources. Tug-of-War measures how models trade parametric and contextual knowledge; FaithfulRAG models fact-level discrepancies; Astute RAG reconciles imperfect retrieval with internal knowledge (Jin et al., 2024b; Zhang et al., 2025b; Wang et al., 2025a). Although all three concern "conflict," their experimental units, source assumptions, and desired responses differ. The appropriate outcome may be answer correctness, contextual faithfulness, calibrated disclosure, or source reconciliation. Earlier entity-substitution experiments and PopQA isolate when parametric memory resists or benefits from retrieved evidence (Longpre et al., 2021; Mallen et al., 2023). ConflictBank broadens the benchmark unit to millions of claim–evidence pairs across misinformation, temporal conflict, and semantic ambiguity (Su et al., 2024); its scale does not make those conflict types interchangeable.

Grounding resources address another layer. RAGTruth provides span-level hallucination labels, while RAGAS and ARES estimate response or context support (Niu et al., 2024; Es et al., 2024; Saad-Falcon et al., 2024). These evaluators can identify an unsupported answer, but support alone does not establish that the cited source is current or authoritative. A system can be faithful to weak evidence or correct for a reason absent from its citations.

Source reliability and time therefore belong in the evidence record. Recent work models source reliability directly (Hwang et al., 2025), and other studies examine source preference and contextual authority (Schuster et al., 2026; Li et al., 2026). The comparison-relevant question is not only whether the model used retrieved text, but whether the benchmark specified which source should govern, at what timestamp, and under what corroboration rule. KILT standardizes provenance against a fixed knowledge source, while FreshQA and RealTimeQA test knowledge whose answer changes over time (Petroni et al., 2021; Vu et al., 2024; Kasai et al., 2023). ALCE and generative-search verifiability work separately score citation attribution and whether cited sources actually support generated statements (Gao et al., 2023; Liu et al., 2023).

Multi-agent methods add yet another intervention boundary. MAIN-RAG uses agent voting to filter retrieved documents, MADAM-RAG assigns agents to conflicting documents before aggregation, and DRAG introduces debate during retrieval and generation (Chang et al., 2025; Wang et al., 2025b; Hu et al., 2025). General multi-agent debate supplies methodological background (Du et al., 2024), but these RAG systems should be compared only when the evidence allocation, agent budget, aggregation rule, and target outcome match.

B.3 What Changes Under an Attacker?

Corpus poisoning makes evidence unreliability intentional. PoisonedRAG inserts passages to steer target answers; earlier corpus-poisoning work manipulates retrieval through adversarial passages (Zou et al., 2025; Zhong et al., 2023). Attack success is conditional on corpus write access, insertion budget, query knowledge, retriever visibility, and target scope. Omitting these fields can make two attack rates numerically similar but scientifically incomparable.

Indirect prompt injection and backdoors target a different interface. Retrieved text may contain instructions that override the user task, or a trigger may activate behavior learned during training

(Chaturvedi et al., 2025; Chaudhari et al., 2024; Cheng et al., 2024). These attacks should not inherit claims from factual poisoning merely because both place malicious text in the context. Retrieval realism, instruction-following behavior, and trigger access define distinct games.

Defenses likewise have different roles. Certified RAG gives guarantees under a bounded corruption model; filtering and semantic isolation demote suspicious documents; traceback identifies likely poisoned sources (Xiang et al., 2024; Si et al., 2025; Zhang et al., 2025a). Prevention, certified accuracy, detection, and recovery require different outcomes. Any cross-defense comparison should pair the same capability envelope with clean utility and state whether the defense prevents, detects, traces, quarantines, or recovers.

B.4 How Does a Signal Become an Action?

Calibration, answerability, and retrieval usefulness are related but not identical signals. Calibration-oriented RAG predicts reliability, whereas Sufficient Context asks whether the retrieved evidence can support an answer (Jang et al., 2025; Joren et al., 2025). A well-calibrated score can still feed a poor threshold or action policy, and an insufficient-context detector does not determine whether retrieval, clarification, or abstention has the highest utility.

Adaptive systems make the policy explicit. Self-RAG learns when to retrieve and critique; CRAG adds corrective retrieval; CtrlA separates uncertainty sources to select control actions (Asai et al., 2024; Yan et al., 2024; Liu et al., 2025b). Their reported gains depend on the available action set and resource budget. Answer-only accuracy cannot capture the benefit or cost of extra retrieval, verification, or refusal.

Selective evaluation should therefore retain risk, coverage, action, and cost rather than collapse them into one score. Trust-or-abstain methods, conformal controllers, and uncertainty benchmarks instantiate different operating points (Zhu et al., 2026; Chen et al., 2024b; Nguyen et al., 2026). Comparisons are strongest when systems share evidence and actions and are evaluated across a risk–coverage or utility curve.

B.5 What Can the Evidence Map Establish?

The map documents that robust-RAG research spans evidence construction, evidence use, adversarial integrity, and action control. Existing surveys provide architectural, evaluation, trustworthiness, security, conflict, and abstention views (Gao et al., 2024; Yu et al., 2024; Zhou et al., 2024; Xu et al., 2026, 2024; Wen et al., 2025). Controlled searches for RAG best practices provide an empirical component comparison under shared tasks (Wang et al., 2024), complementary to survey taxonomies and to the claim-level audit studied here. Table 19 states the narrower claim-level increment tested here.

The map cannot establish that these families occur at representative rates, that its taxonomy is complete, or that one family has stronger evidence. Accepted-paper additions strengthen several historical lines, while 207 of 342 mapped rows remain preprints and coding depth varies across legacy strata. For that reason, map counts provide documentary context only. The main design claim comes from the assignment-unit simulation; the 80-claim human diagnostic is reported separately and does not validate the framework.

Estimator	Exact	κ / p	95% CI
Three-field taxonomy	0.300	-0.003	[0.182, 0.415]
Ablation (no C2/C5)	0.500	0.254	[0.390, 0.594]
Five coordinates	0.550	0.315	[0.494, 0.609]
Five coord.\{ } + controls	0.500	0.273	[0.420, 0.571]
Majority class (valid)†	0.537	0.000	[0.432, 0.646]
Random (marginal)†	0.427	0.000	[0.355, 0.519]
Uniform over 4 verdicts†	0.250	0.000	–
Paired contrasts (source-clustered bootstrap)			
Full − Ablation	+0.000	1.00	[-0.066, +0.079]
5-coord − Ablation	+0.050	0.42	[-0.029, +0.158]
Full − Majority	-0.037	0.76	[-0.163, +0.097]
5-coord − Majority	+0.013	1.00	[-0.099, +0.138]

Table 4: Development diagnostic on 80 comparison claims, with the trivial baselines (†) the schema must beat. Intervals are percentile bootstrap intervals clustered on the source paper. No rule variant separates from the strongest trivial baseline (Majority class (valid), 0.537), and adding the C2 and C5 coordinates to the ablation moves exact agreement by an amount whose interval spans zero. The diagnostic therefore supplies no evidence that the coordinates work as a classifier, which is why the paper's substantive evidence rests on the source-linked comparability cases instead.

C Development Diagnostic Details

Human-validation diagnostic	**Value**
Model-stratum coverage (V/P/I/U)	26/125; 25/134; 25/26; 4/4
Unanimous verdicts	39/80
Adjudicator changes to available majorities	9/78
Pairwise verdict κ	0.258–0.525
C1 / C3 / control-status κ	0.101–0.283 / 0.215–0.535 / 0.208–0.266
Full-rule exact / κ / macro-F1	0.500 / 0.273 / 0.437
Model *invalid* precision	5/25 = 0.200

Table 5: Reliability diagnostics for the 80-claim human sample. V/P/I/U denote the model-valid, partial, invalid, and under-specified strata; unequal coverage and development overlap make these descriptive sample results, not held-out population performance.

Rule	Exact [95% CI]	κ [95% CI]	Macro-F1 [95% CI]
Three-field taxonomy	0.300 [0.182, 0.413]	-0.003 [-0.106, 0.109]	0.180 [0.108, 0.256]
Evidence+controls (C2/C5 ablated)	0.500 [0.389, 0.593]	0.254 [0.079, 0.383]	0.417 [0.247, 0.556]
Five coordinates	0.550 [0.494, 0.612]	0.315 [0.217, 0.393]	0.382 [0.299, 0.449]
Five coordinates + controls	0.500 [0.422, 0.575]	0.273 [0.139, 0.364]	0.437 [0.291, 0.547]

Table 6: Agreement with adjudicated human verdicts on the 80-claim sample. Intervals use 50,000 source-cluster bootstrap resamples over nine sources. The resampling respects within-source dependence but cannot make a non-probability, development-overlapping sample representative. All rows still reuse the same model-coded fields.

Human-adjudicated archetype	n	Sources	Full exact [95% CI]	Full–ablated exact [95% CI]
Self-baseline	12	7	0.333 [0.111, 0.625]	+0.000 [+0.000, +0.000]
Shared harness	22	5	0.500 [0.286, 0.737]	-0.045 [-0.150, +0.000]
Cross-source	19	6	0.474 [0.217, 0.714]	+0.000 [-0.176, +0.214]
Causal/mechanistic	14	5	0.500 [0.154, 0.846]	+0.000 [+0.000, +0.000]
Field-level	5	2	0.600 [0.000, 0.750]	+0.000 [+0.000, +0.000]
Policy/utility	8	5	0.750 [0.556, 1.000]	+0.125 [-0.333, +0.667]

Table 7: Exploratory archetype breakdown with source-cluster intervals. Small strata and few source clusters yield wide or discrete intervals; the rows should not be ranked as stable performance estimates.

Field	Raw agreement	Cohen's κ	Gwet's AC1
Claim archetype	0.637–0.738	0.546–0.659	0.570–0.690
C1 evidence state	0.375–0.550	0.101–0.283	0.256–0.470
C2 origin/capability	0.600–0.738	0.153–0.401	0.549–0.673
C3 intervention	0.537–0.787	0.215–0.535	0.438–0.727
C4 outcome/unit	0.662–0.750	0.253–0.462	0.620–0.718
C5 policy/utility	0.775–0.925	0.445–0.737	0.720–0.913
Control status	0.400–0.537	0.208–0.266	0.273–0.452
Final verdict	0.537–0.787	0.258–0.525	0.434–0.751

Table 8: Ranges over the three human-coder pairs on all 80 claims. Low C1, C3, and control-status agreement localizes the main operational ambiguity.

Field	Exact	κ	AC1
Archetype	0.738	0.670	0.688
C1 evidence state	0.575	0.001	0.532
C2 origin/capability	0.725	0.359	0.679
C3 intervention	0.675	0.302	0.589
C4 outcome/unit	0.762	0.332	0.739
C5 policy/utility	0.887	0.705	0.861
Control status	0.562	0.356	0.474

Table 9: Agreement between model-coded fields and adjudicated human fields on the same 80 claims. This separates field extraction/coding disagreement from the later rule-composition step, although several intermediate rule flags were not human-adjudicated.

Reference labels	n	**Ablated exact**	**Full exact**	**Δ exact**	**$\Delta\kappa$**	**Δ macro-F1**
Adjudicated	80	0.500	0.500	+0.000	+0.019	+0.020
Coder A	80	0.475	0.463	-0.012	+0.007	+0.015
Coder B	80	0.550	0.575	+0.025	+0.046	+0.033
Coder C	80	0.438	0.400	-0.037	-0.020	-0.016
Available majority	78	0.500	0.487	-0.013	+0.006	+0.017
Unanimous subset	39	0.641	0.615	-0.026	-0.030	-0.016

Table 10: Sensitivity of the full-versus-ablated comparison to the reference labels. Individual-coder, majority, and unanimous rows are pre-adjudication diagnostics, not alternative gold standards. The sign of the exact-agreement difference changes across references.

D Comparability Case Ledger

ID	Side A (reported value)	Side B (reported value)	Coordinate-matched reading
SEC-01	PoisonedRAG black-box ASR 0.97 on NQ, GPT-4, 5 injected texts per target question, Contriever	Corpus Poisoning Attack ASR 0.01 on NQ (PoisonedRAG Table 4); original paper reports top-20 retrieval success 80.9% with 50 passages on Contriever-ms	PoisonedRAG ASR at 1 malicious text in top-5 = 0.48 on NQ black-box (Table 17 of poisonedRAG2024.txt L2853-2856)
SEC-02	PoisonedRAG black-box ASR 0.97 on NQ, 5 injected texts per target question, GPT-4, Contriever	Phantom attack success 93.3% for Refusal-to-Answer on netflix trigger, Llama3-8B, Contriever, 1 poisoned document	PoisonedRAG ASR at 1 malicious text in top-5 = 0.48 on NQ black-box (Table 17 L2853-2856); Phantom = 93.3% at 1 document
SEC-03	RobustRAG certified accuracy 38.0% on RQA, Mistral-I7B, k=10, k'=1 malicious passage; empirical ASR <=15%	PoisonedRAG black-box ASR 0.97 on NQ, 5 injected texts, GPT-4, Contriever	RobustRAG: certified accuracy 38.0% at k'=1 (L1303-1306); PoisonedRAG ASR at 1 malicious in top-5 = 0.48 (Table 17 L2853-2856)
SEC-04	BadRAG MCOP 98.2% attack success with 10 adversarial passages; GPT-4 reject ratio 74.6% on NQ (denial-of-service)	PoisonedRAG black-box ASR 0.97 on NQ, GPT-4, 5 injected texts per target question	not_recoverable_from_text
SEC-05	AIP ASR 95.23% on MedSquad, 5 adversarial documents, GPT-3.5-Turbo for rephrasing	PoisonedRAG ASR 28.57% on MedSquad (as reported in AIP Table 1)	Both at 5 injected docs on MedSquad: AIP 95.23% vs PoisonedRAG 28.57%
NOISE-01	Self-CRAG accuracy 61.8% on PopQA, SelfRAG-LLaMA2-7b backbone	Self-RAG 7B accuracy 54.9% on PopQA	Self-CRAG 61.8 vs Self-RAG 54.9 on PopQA (same backbone, retriever, metric)
NOISE-02	CtrlA7B accuracy 61.8% on PopQA, Mistral-7B, BM25+BGE retriever	Self-RAG 7B accuracy 54.9% on PopQA, LLaMA2-7B, Contriever-MS MARCO	not_recoverable_from_text
NOISE-03	ChatGPT accuracy 76.00% at 0.8 noise ratio on English QA subset	ChatGPT3.5 average EM 71.47 under mixed noise on RAG-Bench	not_recoverable_from_text
NOISE-04	Chain-of-Note EM 41.83 vs Retrieve-Read EM 34.28 on NQ at 100% noise ratio, LLaMA-2 7B, DPR	RAAT EM 87.15 vs RALMmultiple EM 85.47 average across noise types, LLaMA2-7B, DPR	not_recoverable_from_text
NOISE-05	ASTUTE RAG accuracy 52.2% on NQ, Claude 3.5 Sonnet, real-world retrieval	RobustRAG keyword aggregation accuracy 47.8% on NQ, Claude 3.5 Sonnet (AstuteRAG Table 1)	ASTUTE RAG 52.2 vs RobustRAG 47.8 on NQ (matched LLM, dataset, metric)
FAITH-01	Finetuned Llama-2-13B response-level hallucination detection overall F1 78.7%	Same models at span-level (Table 6) — different and lower numbers for word-level localization	not_recoverable_from_text
FAITH-02	FaithfulRAG (mistral-7b-instruct) accuracy 81.7% on FaithEval; Self-RAG (Llama2-7B) 37.4% on FaithEval	GPT-4o 47.5% accuracy on counterfactual context task of FaithEval; GPT-4o 96.3% on closed-book QA	not_recoverable_from_text
CALIB-01	Self-RAG 7B accuracy 66.4% on TriviaQA-unfiltered, Contriever-MS MARCO	Llama2-7B with retrieval accuracy 42.5% on TriviaQA-unfiltered	Self-RAG 66.4 vs Llama2-7B RAG 42.5 on TriviaQA (all coordinates matched)
CALIB-02	CtrlA7B accuracy 76.4% on TriviaQA, Mistral-7B, BM25+BGE	FLARE accuracy 72.4% on TriviaQA, reimplemented with Mistral-7B	CtrlA 76.4 vs FLARE 72.4 on TriviaQA (matched backbone, retriever, dataset)
CALIB-03	ADAPT-LLM (SQuAD) accuracy 38.15% on PopQA test set, Llama-2 7B, Contriever, exact match	Self-RAG 7B accuracy 54.9% on PopQA long-tail subset (1,399 queries), Llama2-7B, Contriever-MS MARCO + web search	not_recoverable_from_text

Table 11: Reported values and coordinate-matched readings for every comparability case. The released ledger records the file and line range supporting each value; the verifier re-opens each range and fails on any mismatch.

E Published Human-Evaluation Audit Cases

Reporting field	R	P	NR	N/A
Distinct annotator count	2	0	7	0
Raters per item	2	2	5	0
Item–annotator assignment	0	0	9	0
Aggregation rule	1	2	6	0
Rater agreement/uncertainty	4	0	5	0
Recruitment or qualifications	5	1	3	0
Compensation	2	0	6	1
Disagreement/reference resolution	2	1	6	0

Table 12: Reporting audit of nine published robust-RAG studies that collected original human judgments, identified from 39 venue-linked Map B papers with local full text (19 passed the term screen). R/P/NR denote reported, partly reported, and not reported. None randomized an audit prompt across annotators, so we found no verifiable assignment-unit mismatch. Missing fields are reporting gaps, not evidence of design errors.

Study	Human role	N_r	r/i	Assign.	Agg.	Rater unc.	Resolution
Zou et al. (2025)	author validation of an automatic attack-success metric	–	–	–	–	–	–
Yoran et al. (2024)	manual attribution of model errors to retrieved context	–	–	–	–	–	–
Ming et al. (2025)	human validation and filtering of generated benchmark contexts	–	R	–	R	R	R
Niu et al. (2024)	span- and response-level hallucination annotation	–	R	–	P	R	R
Song et al. (2025)	naturalness evaluation of successful adversarial examples	R	P	–	P	R	–
Huang et al. (2025)	validation of LLM-generated knowledge-conflict labels	R	P	–	–	R	–
Huang et al. (2024)	validation and correction of a generated situated-faithfulness benchmark	–	–	–	–	–	P
Joren et al. (2025)	gold labeling of context sufficiency for autorater evaluation	–	–	–	–	–	–
Liang et al. (2025)	professional construction of adversarial benchmark contexts	–	–	–	–	–	–

Table 13: Case ledger for the published human-evaluation reporting audit. N_r is the distinct annotator count and r/i the number of raters per item; R/P/– denote reported, partly reported, and not reported. The machine-readable audit links each judgment to full-text line ranges. No row assigns a prompt treatment to annotators, and no row is classified as a confirmed assignment-unit error.

F Formal Estimand Definitions

Let reviewer r receive prompt $T_r \in \{0, 1\}$, rate item i, and emit label $Y_{ri}(T_r)$. Let Z_i denote a constructed reference and $s(Y, Z)$ a reference-relative score.

Realized-panel contrast.

$$\Delta_{\text{panel}} = \frac{1}{I} \sum_i \big[s(M_{r \in \mathcal{R}_1} Y_{ri}(1), Z_i) - s(M_{r \in \mathcal{R}_0} Y_{ri}(0), Z_i) \big],$$

where M is a majority rule and $\mathcal{R}_t$ is the realized reviewer panel. This quantity is descriptive for those panels and that reference.

Reviewer-population contrast.

$$\Delta_{\text{rater}} = \mathbb{E}_{r,i,Z}[s(Y_{ri}(1), Z_i) - s(Y_{ri}(0), Z_i)] \,.$$

This target requires treatment assignment and uncertainty at the reviewer level, plus a defined reviewer population and reference distribution.

Consequence contrast.

$$\Delta_{\text{loss}} = \mathbb{E}[L\{a(Y(1)), \theta\} - L\{a(Y(0)), \theta\}] \,,$$

where $a(\cdot)$ is the accept, narrow, reject, or defer decision and θ is the deployment state.

Variance decomposition. Let $\mathcal{R}$ be the realized reviewer composition and $\mathcal{Z}$ the realized reference process. Then

$$\text{Var}(\widehat{\Delta}) = \mathbb{E}\Big[\text{Var}(\widehat{\Delta} \mid \mathcal{R}, \mathcal{Z})\Big] + \text{Var}\Big[\mathbb{E}(\widehat{\Delta} \mid \mathcal{R}, \mathcal{Z})\Big] \,.$$

Source resampling with reviewers and references fixed estimates the first term. Reviewer assignment and reference construction contribute to the second.

Assignment asymptotics for panel majorities. For three conditionally independent votes with correctness probabilities p_1, p_2, p_3, majority correctness has expectation

$$m(p_1, p_2, p_3) = p_1p_2 + p_1p_3 + p_2p_3 - 2p_1p_2p_3.$$

Let Z be a six-versus-six reviewer assignment and $\mu(Z) = \mathbb{E}(\widehat{\Delta} \mid Z)$ the expected difference between the two scheduled panel-majority scores. With I conditionally independent items (or G independent source clusters of bounded size),

$$\text{Var}(\widehat{\Delta}) = O(I^{-1}) + \text{Var}_Z\{\mu(Z)\}.$$

The source-only standard error targets the first term. If reviewer profiles are heterogeneous, the nonlinear function m generally makes $\mu(Z)$ nonconstant, so the second term does not vanish when more items are added to the same twelve assignments. Conditional item precision can therefore converge to zero while uncertainty over reviewer assignment remains. This characterization explains the direction of the simulation; it does not supply a population model for reviewer abilities.

G Assignment-Unit Simulation Details

Quantity	Value
Crossed design	3 coders $\times$ 80 claims
Coder accuracy vs. adjudicated	0.850, 0.775, 0.725
$\hat{\sigma}_r$ (moment estimate)	0.271
Bootstrap 50% interval	[0.233, 0.400]
Bootstrap 95% interval	[0.000, 0.531]
Identification from published reporting	
Widest admissible σ_r set	100% of the grid
Point identified by reporting?	no

Table 14: Reviewer heterogeneity is estimable only from a crossed reviewer-by-item design. The estimate comes from the retained 80-claim diagnostic, bootstrapped over source clusters; three coders make the variance imprecise, so the interval rather than the point is the estimate. The published agreement rates located in the audited papers place no usable restriction on σ_r: at the label prevalence those papers imply, every value on a 0–1.2 grid reproduces the reported agreement for some item dispersion. Separating the two components requires the item-by-annotator assignment, which none of the nine audited papers reports.

Abilities	Quantity	σ_r	G	Claims	Panel/src	Votes/src	Rev. t	Rev. RI
fixed_quantiles	size	0.00	20	120	4.9	5.2	4.9	3.7
fixed_quantiles	power	0.00	20	120	34.7	55.4	50.2	45.0
fixed_quantiles	size	0.00	40	240	4.7	4.9	4.9	4.0
fixed_quantiles	power	0.00	40	240	61.8	85.6	79.1	76.2
fixed_quantiles	size	0.00	80	480	4.9	4.9	4.5	4.0
fixed_quantiles	power	0.00	80	480	90.0	99.2	97.3	96.8
fixed_quantiles	size	0.00	160	960	4.9	4.6	4.2	3.8
fixed_quantiles	power	0.00	160	960	99.6	100.0	100.0	100.0
fixed_quantiles	size	0.23	20	120	8.5	10.9	5.0	3.9
fixed_quantiles	power	0.23	20	120	35.5	54.0	35.0	30.9
fixed_quantiles	size	0.23	40	240	13.3	18.3	4.9	4.4
fixed_quantiles	power	0.23	40	240	59.5	77.8	47.9	45.3
fixed_quantiles	size	0.23	80	480	20.6	28.7	4.7	4.2
fixed_quantiles	power	0.23	80	480	80.5	90.7	58.1	56.3
fixed_quantiles	size	0.23	160	960	31.9	42.2	5.2	4.7
fixed_quantiles	power	0.23	160	960	92.2	96.8	66.1	65.1
fixed_quantiles	size	0.27	20	120	9.7	13.2	4.8	3.9
fixed_quantiles	power	0.27	20	120	36.9	53.9	31.5	28.4
fixed_quantiles	size	0.27	40	240	16.1	22.5	4.8	4.2
fixed_quantiles	power	0.27	40	240	58.9	75.1	42.0	39.7
fixed_quantiles	size	0.27	80	480	25.9	34.5	4.9	4.6
fixed_quantiles	power	0.27	80	480	78.2	88.0	48.7	47.2
fixed_quantiles	size	0.27	160	960	39.1	48.4	5.2	4.8
fixed_quantiles	power	0.27	160	960	90.3	95.1	54.8	54.0
fixed_quantiles	size	0.40	20	120	15.9	21.3	5.0	4.3
fixed_quantiles	power	0.40	20	120	38.1	52.6	21.5	19.1
fixed_quantiles	size	0.40	40	240	26.3	36.0	5.2	4.8
fixed_quantiles	power	0.40	40	240	56.8	70.1	25.5	24.0
fixed_quantiles	size	0.40	80	480	39.8	49.6	4.6	4.1
fixed_quantiles	power	0.40	80	480	73.9	82.1	27.9	26.6
fixed_quantiles	size	0.40	160	960	53.5	62.9	5.3	4.9
fixed_quantiles	power	0.40	160	960	81.9	87.3	27.4	26.8
fixed_quantiles	size	0.53	20	120	23.9	32.2	5.0	4.3
fixed_quantiles	power	0.53	20	120	40.5	53.2	14.5	13.2
fixed_quantiles	size	0.53	40	240	36.7	47.0	5.4	4.9
fixed_quantiles	power	0.53	40	240	56.7	67.0	16.7	15.9
fixed_quantiles	size	0.53	80	480	49.5	58.7	4.8	4.5
fixed_quantiles	power	0.53	80	480	70.0	77.7	16.8	16.1
fixed_quantiles	size	0.53	160	960	64.0	71.7	5.5	5.3
fixed_quantiles	power	0.53	160	960	78.6	83.8	17.7	17.1
resampled	size	0.00	20	120	5.6	5.3	5.2	4.0
resampled	power	0.00	20	120	33.9	53.9	49.4	44.6
resampled	size	0.00	40	240	5.0	5.0	4.8	4.0
resampled	power	0.00	40	240	62.9	86.3	79.7	77.2
resampled	size	0.00	80	480	4.7	5.1	4.7	4.3
resampled	power	0.00	80	480	89.9	99.2	97.5	97.0
resampled	size	0.00	160	960	5.9	5.2	4.6	4.1
resampled	power	0.00	160	960	99.5	100.0	100.0	100.0
resampled	size	0.23	20	120	9.3	11.7	5.1	4.1
resampled	power	0.23	20	120	35.1	52.9	34.6	31.5
resampled	size	0.23	40	240	12.8	16.6	4.5	4.0
resampled	power	0.23	40	240	60.5	77.9	49.8	47.3
resampled	size	0.23	80	480	20.6	27.9	4.9	4.7
resampled	power	0.23	80	480	82.0	91.5	60.4	58.8
resampled	size	0.23	160	960	32.2	41.3	5.0	5.0
resampled	power	0.23	160	960	92.6	96.3	67.8	66.8
resampled	size	0.27	20	120	9.9	13.3	4.8	3.7
resampled	power	0.27	20	120	36.8	53.9	32.1	28.6
resampled	size	0.27	40	240	14.7	21.5	5.0	4.3
resampled	power	0.27	40	240	58.9	75.8	42.7	41.0
resampled	size	0.27	80	480	24.2	32.9	4.6	4.3
resampled	power	0.27	80	480	78.8	89.0	52.8	51.6
resampled	size	0.27	160	960	36.7	46.5	4.4	4.3
resampled	power	0.27	160	960	89.8	94.9	56.3	55.4
resampled	size	0.40	20	120	15.3	21.8	4.9	4.2
resampled	power	0.40	20	120	38.1	54.4	23.3	21.5
resampled	size	0.40	40	240	26.4	34.5	4.8	4.4
resampled	power	0.40	40	240	58.0	70.7	26.5	25.3
resampled	size	0.40	80	480	38.6	49.0	5.7	5.4
resampled	power	0.40	80	480	73.2	81.9	30.9	30.2
resampled	size	0.40	160	960	51.4	60.4	4.3	4.3
resampled	power	0.40	160	960	83.3	88.3	33.1	33.0
resampled	size	0.53	20	120	23.3	30.8	5.3	4.5
resampled	power	0.53	20	120	41.4	54.3	16.7	15.3
resampled	size	0.53	40	240	35.8	45.7	4.7	4.4
resampled	power	0.53	40	240	58.0	68.8	20.0	19.1
resampled	size	0.53	80	480	48.8	57.3	5.1	5.1
resampled	power	0.53	80	480	70.2	78.5	19.9	19.6
resampled	size	0.53	160	960	63.2	70.8	4.7	4.5
resampled	power	0.53	160	960	79.7	85.1	20.5	20.5

Table 15: Complete grid for the retrospective-remedy simulation. *fixed_quantiles* holds reviewer abilities at twelve population quantiles as in the original simulation; *resampled* redraws them each replicate. Entries are rejection percentages.

Setting	Source	Panel RI	Rater RI	SD/SE
$\sigma = 0.20$, $G = 20$	7.6 (0.4)	4.9 (0.3)	3.7 (0.3)	1.14
Homog., $G = 80$	4.9 (0.3)	4.5 (0.3)	4.5 (0.3)	0.99
$\sigma = 0.20$, $G = 80$	17.0 (0.5)	5.3 (0.3)	4.6 (0.3)	1.44
$\sigma = 0.20$, $G = 160$	25.9 (0.6)	4.9 (0.3)	5.2 (0.3)	1.77

Table 16: Oracle design benchmark, retained for reference. Entries are rejection percentages with Monte Carlo standard errors in parentheses, and SD/SE is the empirical standard deviation of the D−C contrast divided by the mean source-only standard error, so a value above one marks an optimistic standard error. Each row uses 5,000 replicates, 12 auditors (six per prompt), six claims per source, and balanced three-of-six panels; G is the number of sources and rater ability follows 12 fixed normal quantiles on the logit scale around 75% mean agreement. Source t and Panel RI use the identical studentized panel-majority statistic; Panel RI refers it to all 924 six-versus-six assignments under a full-vote oracle with an arm-independent schedule, and Rater RI uses individual accuracy. Panel RI is not implementable retrospectively, which is why Table 3 replaces it with two analyses that are, and evaluates them at the estimated σ_r rather than at an assumed one.

Rater profiles	Sources	Claims	Source-only rejection	Panel-RI rejection	Rater-RI rejection	SD/SE
$\sigma_r = 0.20$	20	120	7.6%	4.9%	3.7%	1.14
$\sigma_r = 0.20$	40	240	10.3%	4.5%	4.3%	1.22
Homogeneous	80	480	4.9%	4.5%	4.5%	0.99
$\sigma_r = 0.10$	80	480	8.0%	5.3%	4.5%	1.13
$\sigma_r = 0.20$	80	480	17.0%	5.3%	4.6%	1.44
$\sigma_r = 0.30$	80	480	28.3%	4.8%	4.8%	1.84
$\sigma_r = 0.20$	160	960	25.9%	4.9%	5.2%	1.77

Table 17: Complete null-simulation sweep. Each row uses 5,000 replicates and a true zero prompt effect. Panel RI uses the same studentized panel-majority statistic as the source-only test. A raw-contrast randomization sensitivity and Monte Carlo standard errors are recorded in the released JSON. The four headline rows appear in Table 16.

H Controlled Model-Judge Details

Model	Stratum	Abl.	Plac.	Full	F−P	F−A	Para. SD	McNemar p	Holm p
Qwen2.5-14B	all	60.9	59.0	61.6	+2.6	+0.7	3.3	0.008	–
Qwen2.5-14B	probe competence	94.8	91.9	92.3	+0.4	-2.5	5.5	0.031	–
Qwen2.5-14B	c2 capability	59.9	56.6	61.4	+4.8	+1.5	5.8	0.031	0.000
Qwen2.5-14B	c5 utility	44.9	44.9	46.5	+1.6	+1.6	2.4	0.001	0.068
Qwen2.5-7B	all	60.0	56.0	59.8	+3.7	-0.2	2.4	0.003	–
Qwen2.5-7B	probe competence	89.2	85.4	82.1	-3.3	-7.1	8.8	1.000	–
Qwen2.5-7B	c2 capability	66.1	59.0	70.1	+11.1	+4.0	7.0	0.000	0.000
Qwen2.5-7B	c5 utility	39.3	38.3	38.2	-0.1	-1.0	2.1	0.383	1.000
Qwen3-14B	all	54.1	54.8	61.3	+6.5	+7.2	4.3	0.000	–
Qwen3-14B	probe competence	94.4	96.5	97.3	+0.8	+2.9	2.8	1.000	–
Qwen3-14B	c2 capability	53.5	51.7	65.9	+14.3	+12.4	5.1	0.000	0.000
Qwen3-14B	c5 utility	34.5	37.2	38.8	+1.6	+4.3	5.1	0.002	0.253
Qwen3-8B	all	59.7	60.0	65.9	+5.9	+6.2	4.2	0.000	–
Qwen3-8B	probe competence	69.0	69.2	70.0	+0.8	+1.0	4.0	1.000	–
Qwen3-8B	c2 capability	74.0	74.5	80.5	+6.0	+6.6	7.9	0.016	0.000
Qwen3-8B	c5 utility	40.8	40.9	49.2	+8.2	+8.3	8.6	0.000	0.000
OLMo-2-13B	all	47.4	46.0	47.8	+1.8	+0.4	5.8	0.016	–
OLMo-2-13B	probe competence	69.6	65.0	70.6	+5.6	+1.0	10.5	0.016	–
OLMo-2-13B	c2 capability	50.7	48.3	53.1	+4.8	+2.4	3.1	0.000	0.000
OLMo-2-13B	c5 utility	32.9	34.2	31.0	-3.1	-1.9	9.2	0.021	0.000
OLMo-2-7B	all	41.5	40.2	41.7	+1.4	+0.1	7.8	0.500	–
OLMo-2-7B	probe competence	43.1	48.1	44.8	-3.3	+1.7	4.5	0.500	–
OLMo-2-7B	c2 capability	44.8	46.9	44.3	-2.6	-0.5	5.5	1.000	–
OLMo-2-7B	c5 utility	37.5	29.7	37.5	+7.8	+0.0	15.6	1.000	–
Granite-3.1-8B	all	53.8	54.8	53.7	-1.2	-0.1	2.3	0.011	–
Granite-3.1-8B	probe competence	81.9	85.2	78.1	-7.1	-3.7	7.7	0.003	–
Granite-3.1-8B	c2 capability	55.9	56.4	57.6	+1.2	+1.7	3.5	1.000	0.200
Granite-3.1-8B	c5 utility	37.5	38.1	37.5	-0.6	+0.0	0.7	1.000	0.213
Phi-4-mini	all	61.5	60.8	62.0	+1.2	+0.5	0.7	0.000	–
Phi-4-mini	probe competence	67.3	69.4	69.8	+0.4	+2.5	7.5	1.000	–
Phi-4-mini	c2 capability	75.8	73.2	74.0	+0.7	-1.9	4.7	0.024	1.000
Phi-4-mini	c5 utility	44.2	44.0	46.0	+2.1	+1.9	2.1	0.000	0.213
Falcon3-7B	all	58.4	58.1	59.1	+1.0	+0.7	2.1	0.860	–
Falcon3-7B	probe competence	66.7	66.7	66.7	+0.0	+0.0	0.0	1.000	–
Falcon3-7B	c2 capability	68.8	68.3	69.0	+0.6	+0.2	1.2	1.000	0.901
Falcon3-7B	c5 utility	44.0	43.5	45.5	+2.0	+1.6	4.3	0.678	0.302

Table 18: Per-model, per-stratum results for the version-2 controlled experiment. Accuracies are percentages averaged over four paraphrases; F−P and F−A are the full-minus-placebo and full-minus-ablated contrasts in percentage points; *Para. SD* is the standard deviation of the full-minus-placebo contrast across paraphrases. McNemar tests use per-item paraphrase-majority decisions. Holm adjustment covers gate-passing models on the two substantive strata.

I Framework and Taxonomy Details

Adjacent line	Primary unit	Unresolved comparison question	This paper's increment
General RAG surveys (Gao et al., 2024; Sharma, 2025)	Architectures, modules, applications	Whether similarly named gains concern the same evidence state, outcome, and experimental game	Adds a claim-level comparison lens and an estimand audit; does not claim a venue-complete census.
Evaluation surveys and unified comparisons (Yu et al., 2024; Wang et al., 2024; Alushi et al., 2026)	Metrics, benchmarks, and common-harness reruns	Which controls are required by a specific cross-study or mechanistic inference when rerunning is impossible	Separates scientific coordinates from assignment, participant, item, and reference units; the simulation quantifies one resulting failure mode.
Trustworthy and security RAG surveys (Zhou et al., 2024; Ni et al., 2025; Xu et al., 2026)	Trust dimensions, attack surfaces, and defenses	Whether benign conflict and attacker-created corruption share a transferable capability envelope	Makes origin/access/budget explicit and shows why a prompt-overlapping reference cannot establish independent criterion validity.
Conflict and abstention surveys (Xu et al., 2024; Wen et al., 2025)	Conflict types and refusal behavior	Whether a confidence or sufficiency signal identifies the action that maximizes utility at a matched operating point	Separates signal, action, and utility and releases a consequence-defined counterfactual target without claiming an evaluated policy.

Table 19: Positioning relative to adjacent surveys and empirical comparison work. The increment is a claim-level estimand audit and design stress test, not a new set of primitive concepts or a venue-complete systematic review.

Field	Question	Representative values	Why it changes a comparison
Evidence state	What is observable at the decision boundary?	Supported; distracting; insufficient; conflicting; stale/weak provenance; adversarially manipulated; not evidence-conditioned.	Query ambiguity and model uncertainty are not silently mixed with properties of retrieved evidence.
Origin / capability	How did that state arise, and what can an attacker observe or change?	Benign retrieval; constructed stressor; natural drift/conflict; distribution shift; attacker-created evidence with access, budget, and target.	Transfer depends on the data-generating process and the attacker's capability envelope.
Intervention	What part of the system changes?	Corpus/index; retriever; reranker; context assembly; generator; verifier; system control; evaluation artifact.	Components may be alternatives in a shared harness or complementary stages in a pipeline.
Outcome	What object and unit are measured?	Answer correctness; retrieval quality; claim support; attack success; privacy leakage; calibration; selective risk; cost.	Identically named datasets do not make different target quantities interchangeable.
Policy / utility	What response is selected, under which costs and stakes?	Answer; verify; re-retrieve; expose conflict; clarify; quarantine; recover; abstain; coverage–risk utility.	A signal becomes operational only through an action set and a declared utility context.
Control check	Does the experiment support the intended inference?	Dataset and corpus snapshot; retriever; top-k; generator; decoding; budget; baselines; uncertainty.	Tuple compatibility alone cannot remove experimental confounding.

Table 20: The diagnostic schema. The five fields identify whether two results address a compatible question; a claim-specific control check then determines whether their reported numbers support the intended inference. The fields are not assumed orthogonal or sufficient.

Research line	Evidence condition	Typical intervention	Outcome / policy	Examples
Context construction	Irrelevant or hard negatives; missing support	Retriever/reranker, compression, noise-aware training	Recall, answer accuracy, filter or re-retrieve	IrrelCtx; RAAT; RGB
Evidence use and conflict	Unsupported claims; source or memory conflict; stale evidence	Evidence-reading prompts, conflict modeling, source verification	Claim support, source preference, expose conflict	RAGTruth; FaithfulRAG; Astute RAG
Adversarial integrity	Poisoned corpus, backdoor trigger, retrieved instruction	Corpus governance, isolation, aggregation, certification, traceback	Attack success, clean utility, quarantine or recovery	PoisonedRAG; RobustRAG; security surveys
Uncertainty and control	Insufficient evidence, uncertain answer, variable retrieval need	Calibration, sufficiency estimation, routing/controller	Selective risk, coverage, answer, abstain, clarify	Sufficient Context; CalibRAG; abstention surveys
Evaluation infrastructure	Mixed evidence states and system roles	Benchmark, evaluator, shared end-to-end harness	Multi-objective scorecard and action trace	RAG evaluation surveys; unified comparisons

Table 21: Detailed robust-RAG research landscape. Co-membership in one line identifies a shared concern, not a direct leaderboard: evidence origin, measured unit, policy, and controls still delimit comparison.

Coordinate	Controlled values	Equivalence / boundary rule	Verdict effect
Evidence state	Supported; irrelevant/distracting; insufficient; conflicting; stale/weak provenance; adversarially manipulated; not evidence-conditioned.	Code what is observable at the decision boundary. Query ambiguity, OOD input, confidence, and attacker access are not evidence states by themselves.	A direct performance claim normally requires a common state or an explicit cross-state estimand.
Origin / capability	Benign retrieval; constructed stressor; natural conflict/drift; query or distribution shift; attacker-created evidence; shared clean harness.	Record how the state arose. For security, include corpus write access, query control, retriever visibility, generator access, insertion budget, objective, and target.	Cross-source ranks over different games become partial or invalid; a shared harness may deliberately vary capability as the controlled factor.
Intervention point	Corpus/index; retriever; reranker; context assembly/compression; generator prompting or training; verifier; system-control policy; evaluation-only artifact.	Code the component changed. Different interventions may be compared as alternatives when a shared end-to-end harness controls the remaining pipeline; otherwise they provide component-specific or compositional evidence.	Intervention mismatch alone does not determine the verdict. The intended claim and shared controls decide whether the comparison is valid or partial.
Outcome / unit	Answer correctness; retrieval quality; evidence support; attack or defense utility; privacy leakage; calibration; selective risk/coverage; cost/latency; human preference.	Record definition, direction, aggregation, and unit. Relative gains over different baselines are not automatically a shared estimand.	A one-scalar ranking requires a common target; otherwise report a qualified multi-objective comparison.
Policy / utility	Answer; abstain; re-retrieve; filter; verify; expose conflict; clarify; quarantine; recover; not applicable; unreported.	Use *not applicable* for fixed descriptive comparisons. When policies compete, report action set, stakes, preferences, coverage, and cost.	Different actions can be ranked only under common utility; a signal alone does not establish the preferred response.
Second-stage controls	Dataset/corpus snapshot; retriever/top-k/chunking; generator/decoding; compute budget; baselines; uncertainty/statistical design.	Select controls according to the stated inference. The tuple cannot remove confounding from unmatched experimental systems.	Tuple compatibility is necessary diagnostic information but not sufficient for a direct empirical ranking.
Final verdict	Valid; partial; invalid; under-specified.	Valid = semantic fields and relevant controls support the stated claim. Partial = only a component-level or qualified claim is supported. Invalid = assumptions or objectives define a different game. Under-specified = a decision-critical field remains unavailable after full-text inspection.	Verdicts attach to comparison claims, not permanently to paper pairs. A new shared-harness experiment can change the supported verdict.

Table 22: Two-stage decision procedure. The five coordinates diagnose semantic compatibility; claim-specific experimental controls determine whether the claim as written is warranted.

Boundary case	Five-coordinate diagnosis	Second-stage control check	Supported conclusion
Different interventions, shared harness	A retriever filter and a generator-side reader address the same evidence state, metric target, and answer policy; intervention differs.	Same corpus, retriever output before intervention, generator, budget, metric, and seeds.	A valid end-to-end comparison is possible. Intervention mismatch alone does not force *partial*.
All five fields match, systems differ	Two answer-only methods target benign retrieval noise and answer correctness at the generator.	Different datasets, retrievers, model scales, top-k, and compute budgets.	The papers are conceptually comparable but their reported numbers are not directly rankable without a shared rerun or adjustment.
Action unnamed but observable	A QA benchmark never uses "action policy" but requires one answer for every item and gives no abstention channel.	Evaluation code and output schema confirm answer-only behavior.	Code *answer* when policy is decision-relevant; use *not applicable* for a fixed descriptive comparison.
Same pair, different claim	An attack artifact and a defense share a threat model but optimize attack success and defended utility.	A common attack–defense harness can evaluate both roles; a paper-only scalar comparison cannot.	Valid as components of one evaluation, partial as substitutable methods, invalid if ranked under incompatible objectives.
Same label, incompatible capability	Two poisoning studies both report attack success, but one assumes bounded corpus insertion and the other controls retrieved instructions at query time.	Capability, budget, retrieval probability, and objective do not align.	A direct ranking is invalid even if intervention, metric name, and answer behavior appear similar.

Table 23: Counterexamples delimiting the schema. Verdicts depend on the stated comparison and experimental controls; the five coordinates are diagnostic fields rather than a necessary-and-sufficient classifier.

Boundary case	Rule used in this paper	Verdict consequence	Example implication
Noisy retrieval vs. distraction	Retrieval imperfection alone is coded as relevance/noise; harmful influence on generation, confidence, or reasoning is coded as distraction.	Same broad topic but different measured effect is partial.	Random negatives and hard distractors should not be pooled into one score.
Query/OOD vs. evidence state	Query ambiguity, distribution shift, and model uncertainty describe the need or belief state. Code them under origin unless the study labels the resulting retrieved evidence.	Shared uncertainty language does not imply a common evidence state or policy.	An OOD detector is not automatically an insufficient-context detector.
Natural conflict vs. adversarial corruption	Code by the source of contradiction, not by surface form. Naturally occurring or diagnostic contradiction is conflict; attacker-created contradiction is integrity/adversarial corruption.	Natural-conflict and poisoning results are invalid as a direct ranking unless the threat model is redefined.	Faithfulness or source-preference methods are not poisoning defenses by default.
Intervention mismatch	Code the component changed, then inspect the evaluation harness. Different interventions can be compared as alternatives if the remaining pipeline and target are controlled.	Shared harness and end-to-end target can support valid comparison; unmatched pipelines support only a qualified comparison.	A filter and a note-taking prompt can be compared after fixing retriever, generator, evidence states, budget, and action utility.
Metric overlap	Metric targets must match the object being measured. Accuracy, citation support, attack success, calibration, and selective risk are separate families.	Shared dataset but different metric family is partial; mutually incompatible targets are invalid for one-scalar ranking.	A correct answer with unsupported citation should not pass a faithfulness comparison.
Observable but unnamed action	Infer behavior only when action is decision-relevant and the protocol is unambiguous; otherwise use *not applicable*.	Merely omitting the phrase "action policy" does not force under-specification.	An answer-only benchmark supplies *answer* for a policy claim, but a fixed latency comparison needs no action label.
Security capability missing	Mark only decision-critical missing access, budget, objective, or target as unknown. A shared harness may intentionally vary capability.	Unknown capability makes a cross-game rank under-specified; controlled capability variation can support a valid descriptive contrast.	Black- and white-box variants of one attack can be compared inside the same harness without implying cross-paper transfer.
Benchmark vs. method	Code benchmark if the reusable artifact is the main contribution; otherwise code the intervention.	Benchmark-method comparison is partial unless the method is evaluated on the benchmark under aligned assumptions.	RGB is benchmark-first; RAAT is training-first.
Tuple match but control mismatch	After semantic coding, inspect corpus, data, retriever, generator, top-k, budget, baselines, decoding, and uncertainty.	A tuple match is insufficient for direct ranking when these factors confound the reported score.	Two answer-only noise methods on different datasets and generators require a shared rerun or a qualified claim.

Table 24: Boundary rules for applying the revised diagnostic schema. Verdicts are claim-specific and require both semantic and experimental-control checks.

Threat class	Corpus	Query	Retr.	Gen.	Objective	Boundary
Dense-retriever poisoning (Zhong et al., 2023)	write	train set	white/gray	no	retrieve bad passages	Not always end-to-end.
RAG knowledge corruption (Zou et al., 2025; Xue et al., 2024)	write	target topic	black/gray	black	answer steering	Stealth/dynamics vary.
Backdoor-style RAG (Cheng et al., 2024; Chaudhari et al., 2024)	write	trigger	gray	black	latent behavior	Normal queries may pass.
Indirect prompt injection (Chaturvedi et al., 2025; Chang et al., 2026)	write/share	natural	black	black	instruction hijack	Must overcome retrieval.
Opinion/fairness manipulation (Chen et al., 2025b; Bagwe et al., 2025)	write	group/topic	black/gray	black	bias steering	QA metrics miss social harm.
Defense setting (Xiang et al., 2024; Yin et al., 2026; Choudhary et al., 2025; Zhang et al., 2025a)	bounded	varies	assumed	black/gray	detect/demote/trace	Depends on budget/visibility.

Table 25: Attacker-capability matrix for adversarial RAG. The same "attack success rate" can mean different things depending on corpus write access, query control, retriever visibility, generator access, and objective.

J Evaluation and Reporting

Required field	What to report	If omitted	Example consequence
Evidence-state definition	Supported, irrelevant/distracting, insufficient, conflicting, stale/weak provenance, adversarially manipulated, or not evidence-conditioned.	Papers with the same robustness label may test different evidence properties.	Noise training is ranked against sufficiency prediction.
Origin / capability	Benign retrieval, constructed stressor, natural conflict/drift, distribution shift, or attacker access, budget, objective, and target.	Stressors are not portable.	Natural conflict is compared with poisoning.
Intervention boundary	Corpus/KB, retriever/context, context assembly, generator, system control, or evaluation artifact.	Pipeline modules are treated as substitutable.	A benchmark is ranked against a mitigation method.
Retrieval setting	Corpus snapshot, retriever, top-k, chunking, reranking/compression, and index drift controls.	Results cannot be reproduced or normalized.	A filter looks better because its retriever retrieved easier negatives.
Outcome and unit	Answer correctness, support, attack success, privacy leakage, selective risk, calibration, cost, or action utility, including definition and aggregation.	Different objectives collapse into one score.	Lower attack success hides clean-utility loss.
Policy and utility	Answer, abstain, retrieve again, filter, verify, expose conflict, clarify, quarantine, or recover, plus stakes, preferences, coverage, and cost when actions compete.	Accuracy and confidence are over-interpreted.	A calibrated answerer may still answer from insufficient evidence.
Model and comparison controls	Dataset, generator, decoding, compute budget, baselines, seeds, uncertainty intervals, and significance procedure.	Tuple-aligned papers can still be empirically confounded.	A method appears better because it uses a stronger generator or a larger budget.
Uncertainty / abstention handling	Coverage, selective risk, confidence target, abstention reward, and re-retrieval cost.	Abstention can hide retrieval failure.	Refusal is rewarded when re-retrieval should be the correct action.
Security threat model	Attacker access, query control, retriever visibility, insertion budget, objective, and adaptive-defense assumptions.	Security numbers are not comparable.	Prompt injection defenses are claimed to protect against corpus poisoning.

Table 26: Reporting checklist for robust RAG under unreliable evidence. The five diagnostic fields identify the comparison question; claim-specific controls are additionally required for empirical inference.

Benchmark	Risk focus	Task / data	Perturbation or threat	Metrics	Boundary
RGB (Chen et al., 2024a)	noise, rejection, integration, counterfactuality	English/Chinese QA	noisy passages, false evidence, negative evidence	accuracy, ability-specific scores	Broad diagnostic benchmark but not adaptive attack-oriented.
RAGTruth (Niu et al., 2024)	hallucination / faithfulness	QA, data-to-text, summarization	naturally generated unsupported spans	word-level hallucination labels	Strong annotation granularity; less focused on retrieval-side threat models.
Sufficient Context (Joren et al., 2025)	answerability	QA with context sufficiency labels	insufficient retrieved evidence	selective generation / abstention	Captures answerability but not adversarial corruption.
MagicMushroom (Zhang et al., 2025c)	retrieval noise	configurable RAG stress tests	distracting, low-quality, inconsequential, irrelevant noise	robustness under noise types	Fine-grained noise; limited conflict/security coverage.
URAG (Nguyen et al., 2026)	uncertainty	RAG as MCQA / conformal prediction	irrelevant/noisy retrieval and uncertainty sets	coverage, set size, calibration	Good for uncertainty; less direct for free-form generation.
QueryRobust (Percin et al., 2025)	query robustness	QA with query perturbations	wording variation at query–retrieval interface	retrieval and end-task degradation	Focuses on query perturbation, not corpus corruption.
PoisonArena (Chen et al., 2025a)	adversarial poisoning	RAG attack competition setting	competing poisoning attacks	attack success, retrieval effects	Strong security focus; benign uncertainty is secondary.
SafeRAG (Liang et al., 2025)	RAG security	security benchmark for RAG	poisoning / malicious retrieval contexts	attack and defense metrics	Security-oriented; not a full evidence reliability suite.
MTRAG-UN (Rosenthal et al., 2026)	multi-turn unanswerability	multi-turn RAG conversations	unanswerable, underspecified, non-standalone questions	abstention / clarification quality	Covers interaction; less focused on adversarial evidence.
CRUMQs (Liu et al., 2025a)	unanswerable multi-hop RAG	realistic multi-hop queries	unanswerable and uncheatable questions	answerability and robustness	Useful for abstention; not a broad attack/noise benchmark.

Table 27: Benchmark-level comparison for robust RAG evaluation. The table makes explicit what each benchmark perturbs, what it measures, and which failure modes remain outside its scope.

Benchmark component	Required annotation	Metric suite	Why it is needed
Evidence reliability	relevance, sufficiency, freshness, provenance, support span	retrieval recall, sufficient-context accuracy, support precision	Distinguishes merely related passages from answer-supporting evidence.
Distraction and conflict	distractor type, conflicting claim pair, source agreement	answer flip rate, conflict sensitivity, context-utilization score	Reveals whether models ignore harmful context or silently pick a source.
Adversarial corruption	attacker access, insertion budget, trigger type, poisoned document ID	attack success, poisoned retrieval rate, stealth, defense bypass	Makes security results comparable across threat models.
Unanswerability and ambiguity	answerability label, ambiguity type, clarification target	selective risk, abstention precision/recall, clarification accuracy	Evaluates robust action selection, not just answer generation.
Cost and system action	number of retrievals, verifier calls, latency, final action trace	cost-adjusted accuracy, action utility, coverage–risk curve	Prevents methods from hiding robustness gains behind unbounded computation.

Table 28: A proposed benchmark blueprint for robust RAG under unreliable evidence. This is a research agenda rather than a validated leaderboard; existing benchmarks cover parts of the schema.

Axis	Typical variation across papers	Why conclusions fail to transfer	Minimum reporting requirement
Retriever/generator setting	BM25 vs dense vs hybrid retrieval; small vs frontier generators; frozen vs fine-tuned models	A defense may work by compensating for one retriever or one model scale rather than solving the failure mode.	Retriever, index, top-k, reranker, generator, prompt, decoding, and context length.
Attack budget and access	Poisoned-document count, corpus write access, query control, retriever visibility, generator access	Attack success rates are incomparable when one attacker edits a corpus and another only controls the query.	Capability matrix, insertion budget, target set, stealth objective, and whether malicious evidence is retrieved.
Metric target	EM/F1, faithfulness, citation support, attack success, calibration error, selective risk, cost	Improving answer accuracy can worsen citation faithfulness; improving abstention can reduce coverage; reducing attacks can lower utility.	Report utility, evidence support, risk, calibration, cost, and security as a Pareto surface.
Statistical and reproducibility evidence	Single seed/model vs multiple seeds/models; code/data availability varies	Small prompt or model changes can reverse robustness rankings, especially for LLM-as-judge or attack benchmarks.	Seeds, confidence intervals or bootstrap tests where feasible, released prompts, data splits, and code or exact API settings.
Long-context baseline	RAG-only comparisons vs long-context LLMs with the same documents	A robustness gain may reflect retrieval compression rather than evidence reliability control.	Include a long-context or full-context baseline when context length permits, and report latency/cost.
Attribution vs answer accuracy	Some tasks score only final answer; others require evidence-supported citations	A model can answer correctly for the wrong reason, cite irrelevant evidence, or ignore retrieved support.	Claim-level support labels, citation precision/recall, and answer correctness reported separately.

Table 29: Evaluation-assumption matrix for robust RAG. The table summarizes why benchmark results often cannot be compared without aligning system setup, threat model, metric target, reproducibility evidence, and long-context baselines.

Main claim	Cited evidence basis	What the evidence supports	Boundary
Irrelevant or noisy context can harm generation.	Irrelevant-context robustness, RAAT, RGB, and counterfactual-noise studies.	Noise affects both retrieval quality and generator context use.	Does not imply one robustness score covers all noise types.
Knowledge conflict differs from adversarial corruption.	Conflict and poisoning studies with different evidence origins and evaluation targets.	Natural disagreement and attacker-created contradiction require different assumptions.	Conflict-resolution results should not be read as poisoning-defense results.
RAG attacks require capability normalization.	Poisoning, backdoor, prompt-injection, traceback, and certification studies.	Attack/defense results require corpus access, query control, retriever visibility, and budget.	The paper does not rank attacks by real-world prevalence.
Calibration, abstention, and retrieval control are distinct.	Sufficient-context, uncertainty, adaptive-retrieval, and calibration studies.	Confidence estimation, evidence sufficiency, retrieval triggering, and abstention optimize different quantities.	Does not validate a single universal control policy.
Action-trace evaluation is a benchmark agenda.	Benchmarks exposing subsets of answer, evidence, action, cost, and threat fields.	Existing benchmarks cover pieces of the proposed schema.	No new leaderboard or empirical system ranking is claimed.

Table 30: Narrative claim-to-source ledger. Expanded and strict citation audits were model-assisted and are not human validation; this table records the intended evidence boundary rather than certifying every citation.

K Scoping-Map Audit Artifacts

Evidence layer	Unit and size	Structured coverage	Supported use and provenance boundary
Scoping corpus	536 broadly included papers	Metadata and broad relevance	Documentary recall pool; single model-assisted broad screen, not duplicated and not prevalence-estimating.
Map B	342 paper rows	Literature family and intervention; free-text metrics; no structured evidence origin or policy	Source-stratified landscape synthesis; adjudicated subsets, but not a five-coordinate map.
Claim inventory	289 claims from purposive fixed sections	Model-coded coordinates, evidence flags, and claim-specific controls	Source-linked hypothesis index; its verdict distribution and internal ablations are not validated results.
Reliability study	80 model-stratified claims	Three human coders, one adjudicated label, and model-rule baselines	Negative stress test; unequal and partly unrecoverable sampling precludes population performance estimates.
Excluded protocol	Planned 480 claims from 80 paper-disjoint sources	Four reviewer-prompt arms and a separate reference process	Design-audit case only. No response, reference statistic, participant attribute, or timing result is accepted as human evidence; retained outcome files document the exclusion trail rather than a study result.

Table 31: The released evidence layers answer different questions. Larger model-assisted layers do not validate claim-level verdicts; the human layers test an annotation procedure rather than literature prevalence. No layer is treated as an independently validated gold standard.

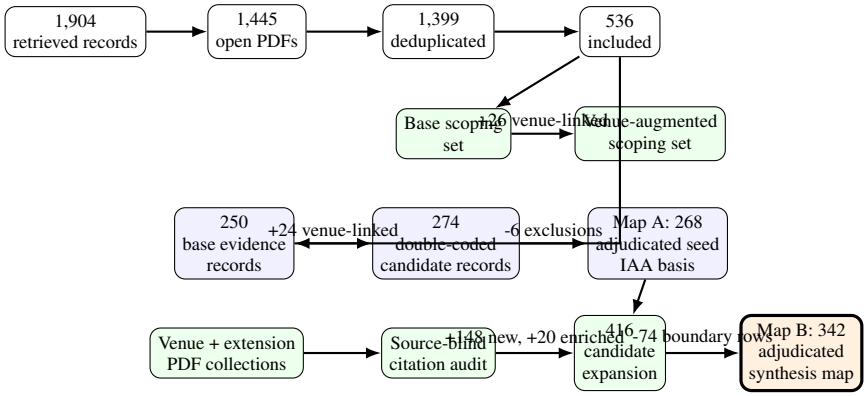


Figure 3: Corpus lineage. The 416-row venue-expanded candidate map is second-coded and adjudicated into Map B (342 rows), the canonical synthesis evidence map. Map A (268 rows) is the double-coded adjudicated seed used for IAA and adjudication audit. The base and venue-augmented scoping sets are used for boundary checks, while a separate 300-paper source-blind citation audit over the venue and extension PDF collections decides which papers enter the main text, case studies, or appendix.

Stage	Count	Rule
Initial retrieval	1,904 records	Keyword search over arXiv, ACL Anthology, OpenReview, USENIX, AAAI/ACM landing pages where open versions were available, plus curated robust/trustworthy RAG paper lists; search cutoff 2026-06-03.
Open full-text corpus	1,445 PDFs	Downloaded only open PDFs; removed failed downloads, inaccessible files, and non-PDF artifacts.
Deduplicated candidates	1,399 papers	Deduplicated by normalized title and arXiv/proceedings identifiers; retained the most complete metadata record when preprint and proceedings versions overlapped.
Relevance screening	536 included papers	Included papers that study RAG robustness, reliability, uncertainty, calibration, faithfulness, attack/defense, or evaluation under unreliable evidence; excluded application-only RAG papers.
Core pool	112 papers	Papers directly shaping the taxonomy, threat/failure model, intervention mechanisms, or evaluation framework.
Core reading set	100 papers	Core papers used as main narrative anchors; the remaining 12 core-pool papers stay in the broader scoping set.
Base scoping set	300 papers	Core papers plus representative supporting work across all taxonomy branches before venue follow-up triage.
Venue-augmented scoping set	326 papers	Base scoping set plus 26 venue-linked additions after follow-up triage.
Candidate evidence records	274 papers	Base evidence map of 250 papers plus 24 venue-linked additions, double-coded and adjudicated.
Map A adjudicated seed	268 papers	Candidate evidence records retained after final adjudication; used for IAA/adjudication provenance.
Map-B candidate expansion	416 papers	Map A plus 148 source-blind human-citation-audit additions and 20 enriched original rows; used for second-coding review.
Map B adjudicated synthesis map	342 papers	Map A plus 74 retained additions after second coding and final adjudication; frozen for current descriptive statistics. Post-freeze discoveries enter a recall log until a new version is coded.

Table 32: Screening-flow summary. Search ends on 2026-06-03; coding and adjudication artifacts freeze on 2026-06-15. The supplement provides query strings, source counts, rules, exclusions, hashes, and a post-freeze recall log.

Map-B field group	Coverage	Coding / audit source	Use and caveat
Canonical identifiers and source flags	342/342 title, year, URL, source stratum, Map-A membership, Map-B version, core-reading flag	Deterministic merge from Map A, the venue/extension pools manifests, source-blind citation-audit decisions, and second-coding adjudication.	Defines the canonical adjudicated row set and source strata.
Core taxonomy fields	342/342 legacy literature-family tag, taxonomy categories, method family, intervention point, contribution note, limitation note	Map-A adjudicated labels for original rows; second coding adjudicated labels for retained additions.	Used for Map-B description. The legacy family tag includes calibration, benchmark, and survey roles and is not the normalized failure coordinate.
Task / metric metadata	Task metadata present; metric field is free text and missing for 13/342 rows	Map-A labels plus audit rationales and manifest metadata for retained additions; no controlled-coordinate reliability estimate.	Used only as descriptive metadata, not as an adjudicated metric coordinate.
Evidence assumption and action policy	**Not present as structured Map-B fields**	Fully instantiated only in worked comparison and pair-study artifacts.	Map B must not be described as a 342-row five-coordinate map or used to derive pair verdicts on these fields.
Author / venue metadata	Source/venue strata complete; author metadata incomplete for some additions	Proceedings metadata and paper manifests.	Venue/source strata are complete; author normalization remains an artifact-release cleanup item.
Human-audited rows retained in Map B	94 retained rows with human citation-audit metadata	74 retained additions plus 20 enriched Map-A rows.	Citation-audit metadata supports source-stratified synthesis; 74 excluded candidate additions remain in the release as boundary rows.
Second-coding/adjudication boundary	Map A: 274 candidates $\rightarrow$ 268 adjudicated rows; Map-B additions: 148 candidates $\rightarrow$ 74 retained additions	Map A has two-source independent coding and final adjudication; Map-B additions have independent second coding and final adjudication.	Second-coding agreement is strong, and adjudication is now applied to the canonical 342-row Map B.

Table 33: Coding protocol and field coverage for the 342-row adjudicated Map B. The map supports literature-family/intervention landscape analysis; it does not implement the complete five-coordinate schema.

Artifact / set	Count	Count definition	Use in this paper
Map-A double-coded candidates	274	Candidate records coded by two human sources before final inclusion adjudication.	Used for human-human agreement and adjudication; six records were excluded from Map A.
Map A adjudicated seed	268	Included records after adjudication: 42 venue/proceedings-linked, 222 arXiv-only or emerging preprints, 2 preprints with year tags, 1 review-platform record, and 1 other/uncategorized record.	Used as the IAA/adjudication provenance snapshot, not the current synthesis denominator.
Venue-augmented scoping set	326	Original base scoping set plus 26 triaged venue-linked additions.	Used for broader scoping and boundary checks; not all rows are evidence-map records.
Source-blind citation audit	300	Human citation decisions over venue PDFs and extension PDFs.	Determines which audited papers enter the main text, case studies, appendix, background, or exclusion set; distinct from the base scoping set.
Map-B candidate expansion	416	Map A plus 148 candidate additions supported by the source-blind citation audit; 20 original rows enriched with citation-audit metadata.	Pre-adjudication expansion candidate map used for second-coding review.
Map B adjudicated synthesis map	342	Map A plus 74 retained additions after second coding and final adjudication.	Current evidence map for source-stratified synthesis and Map-B cross-tabs.

Table 34: Reconciliation of reported corpus counts. The 342-row adjudicated Map B is the canonical synthesis map; Map A and the 274-candidate set are retained to report double-coding and adjudication provenance.